

Chapter-1: Discrete-Time Signals and Systems

❑ Discrete-Time Signal, Basic Signal Types:

1. Signal Definition:

- A **signal** is defined as any physical quantity that varies with time ,space or any other independent variable or variables. A signal carry information. Mathematically, we describe a signal as a function of one or more independent variables.

- For example,

$$x(t) = 10t ; \text{ signal varies linearly with time } t$$

$$x[n] = n^2 + 10n + 15; \text{ signal varies quadratically with time } n$$

$$I(x, y) = 10x^2 + 20xy + 30y^2, \text{ (image signal)}$$

- A signal can be either **real-valued** or **complex-valued**. Independent variable can be either **continuous-time** or **discrete-time**.

2. Signal Classification:

- Signals can be classified on the following two basis:

- A. On the basis of number of independent variables, and
- B. On the basis of function

A. On the basis of number of independent variables:

➤ On this basis , signals can further be classified as

1) One dimensional signal:

➤ It is the function of only one independent variable.

2) Multi-dimensional signal:

➤ It is the function of more than one independent variable.

B. On the basis of function:

➤ On this basis, signals can be further classified as:

1) Continuous-time(CT) and Discrete-time(DT) signals

2) Even(symmetric) and odd(asymmetric or anti-symmetric) signals

3) Periodic and aperiodic signals

4) Energy and power signals

5) Deterministic and random signals

1) Continuous-time(CT) and Discrete-time(DT) signals:

○ Continuous-time(CT) signals:

- A continuous-time(CT) signal is a signal where the value of the function is continuous (real and/or complex values) over all the period of time t . It is denoted by $x(t)$. That is, $x(t)$ is defined for all the time t .

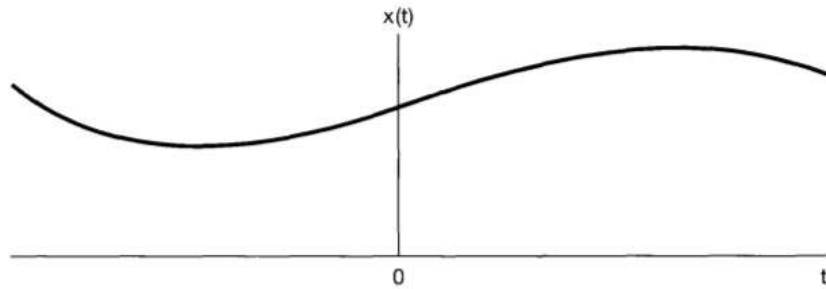


Fig1: Example of CT signal

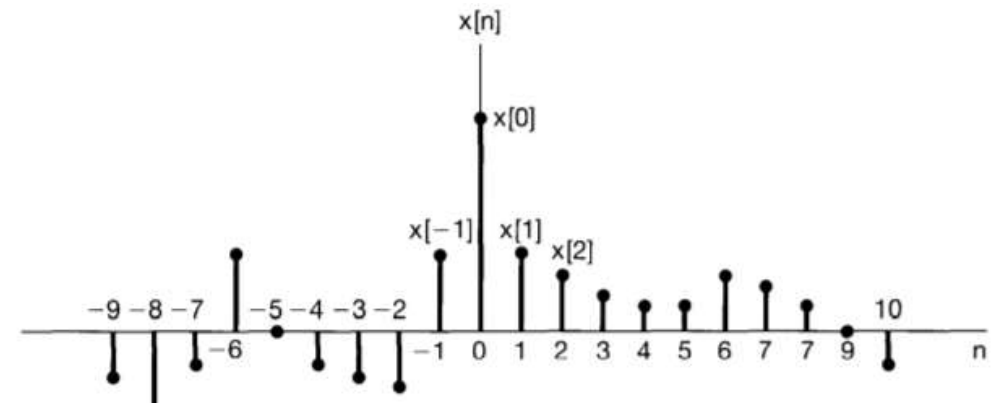


Fig2: Example of DT signal

○ Discrete-time signals:

- A discrete-time (DT) signal is a signal which is defined only at discrete instants of time which are integers. Thus the independent variables have uniformly spaced discrete values only. It is denoted by $x[n]$.

■ **Note:**

- A DT signal between two successive samples is not defined.
- A DT signal can be obtained from a CT signal by sampling it at a uniform rate.

2) Even (symmetric) and odd (asymmetric or anti-symmetric) signals:

○ **Even signals:**

- A discrete-time (DT) signal is referred to as an even signal if it is symmetrical about the vertical axis (i.e., origin). Mathematically,

$$x[-n] = x[n] \text{ for all } n.$$

It is denoted as $x_e[n]$.

○ **Odd Signals:**

- A discrete-time (DT) signal is referred to as an odd signal if it is asymmetrical about the vertical axis (i.e., origin). Mathematically,

$$x[-n] = -x[n] \text{ for all } n.$$

It is denoted as $x_o[n]$.

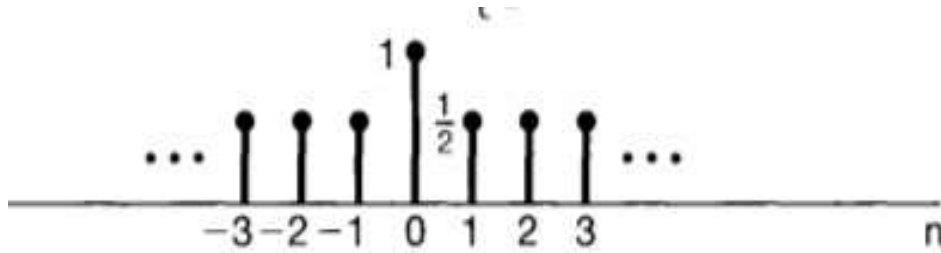
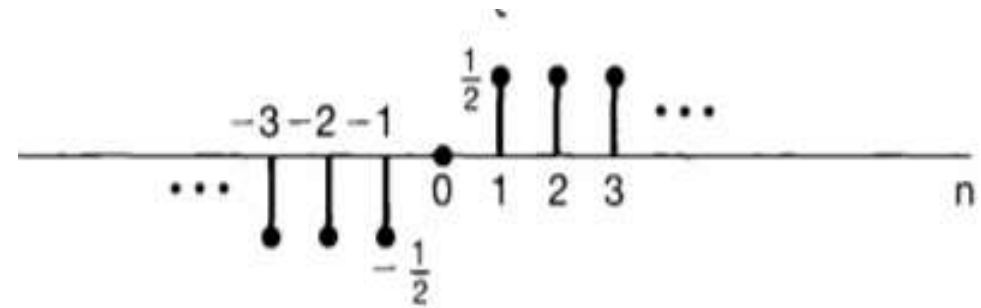


Fig.3: a) A DT even signal



b) A DT odd signal

■ Even and odd components of a Signal:

- Every signal can be expressed(i.e., broken down) as a sum of even and odd components(or signals).That is,

$$x[n] = x_e[n] + x_o[n]$$

then we get

$$x_e[n] = \frac{1}{2} \{x[n] + x[-n]\}$$

and

$$x_o[n] = \frac{1}{2} \{x[n] - x[-n]\}$$

Proof:

➤ We have, $x[n] = x_e[n] + x_o[n]$ i

Put $n=-n$, we obtain

$$x[-n] = x_e[-n] + x_o[-n]$$

$$x[-n] = x_e[n] - x_o[n] \quad \text{.....ii}$$

➤ Adding (i) and (ii), we get

$$x[n] + x[-n] = 2 x_e[n]$$

Therefore, $x_e[n] = \frac{1}{2} \{x[n] + x[-n]\}$

➤ Again subtracting (ii) from (i), we get

$$x[n] - x[-n] = 2 x_o[n]$$

Therefore, $x_o[n] = \frac{1}{2} \{x[n] - x[-n]\}$

3) Periodic and Aperiodic Signal:

➤ A DT signal $x[n]$, it is periodic for some positive constant N if

$$x[n + N] = x[n] \text{ for all } n.$$

where N is the period of the signal.

The smallest value of N that satisfies above equation is called fundamental period.

$$N = \frac{2\pi}{\omega_0} \text{ and } \omega_0 = \frac{2\pi}{N}$$

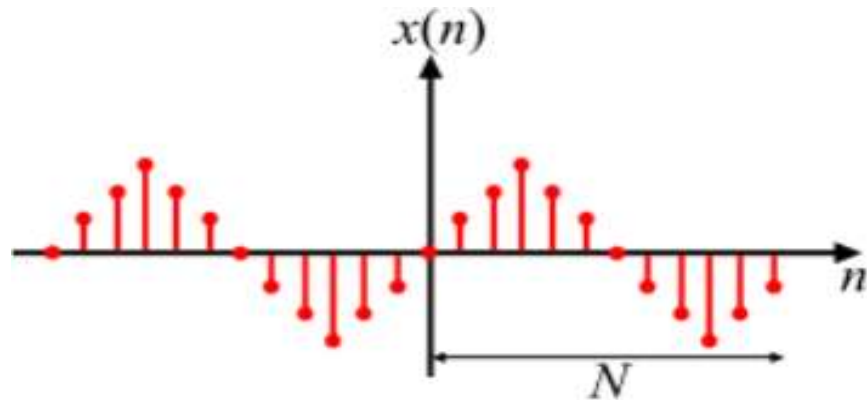


Fig1: A DT periodic signal

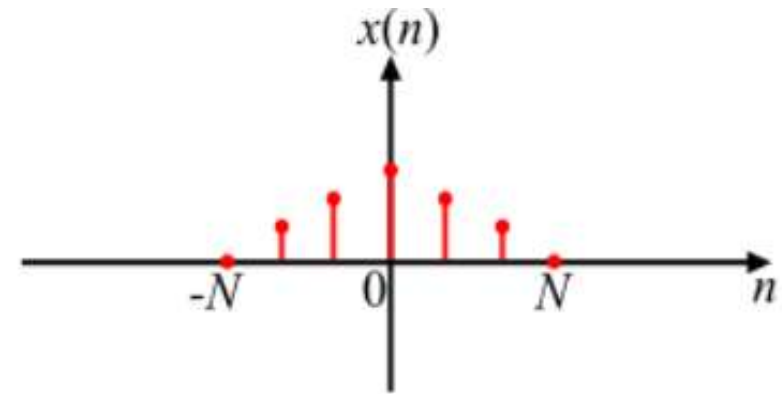


Fig2: A DT aperiodic signal

4) Energy and Power Signal:

- **Energy signal definition:**

- A signal (CT or DT) is said to be an energy signal if the energy of that signal is finite and average power is zero. Mathematically,

$$0 < E < \infty \text{ and } P = 0$$

where, E = signal energy , and P = signal average power

- **Power signal definition:**

- A signal (CT or DT) is said to be an power signal if the average power of that signal is finite and energy is infinite. Mathematically,

$$0 < P < \infty \text{ and } E = \infty$$

where, E = signal energy , and P = signal average power

- Note that if the signal does not satisfy any of the above conditions ,it is neither an energy signal nor a power signal. For example ramp signal.

○ **Signal Energy:**

➤ For discrete-time signal $x[n]$ the total energy is given as

$$E = \sum_{n=-\infty}^{\infty} x[n]^2, \text{ for real-valued signal}$$

$$E = \lim_{N \rightarrow \infty} \sum_{n=-N}^N x[n]^2$$

For complex-valued signal

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$E = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2$$

○ **Signal Power:**

➤ For discrete-time signal $x[n]$, the power is

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x^2[n], \text{ for real signal}$$

For complex-valued signal

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

➤ From above equations, we observe that

$$P = \frac{1}{\text{Period}} E$$

5) Deterministic and Random Signals:

○ **Deterministic signals:**

- A signal whose description is known completely , in either a mathematical form or a graphical form, is called a deterministic signal.
- For example,

$$x[n] = A \cos \omega_0 n$$

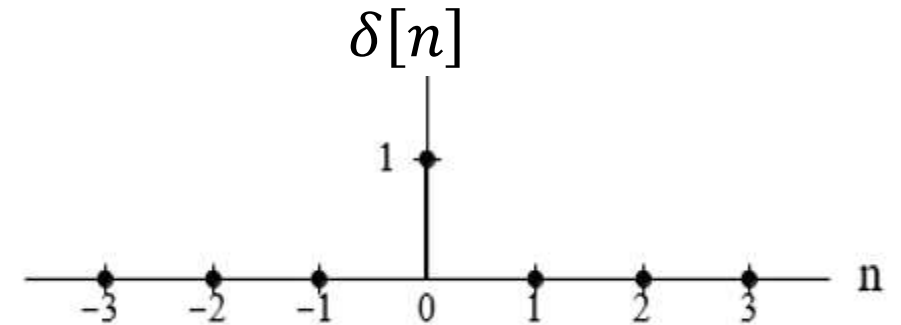
○ **Random signal:**

- If a signal is known in terms of probabilistic description, such as mean value, mean square value and so on, rather than its complete mathematical or graphical description, is called a random signal.
- For example, most of the noise signals in communication systems are random signals.

3. Basic Signal Types:

➤ The basic signals are:

- a) Unit Impulse Signal
- b) Unit Step Signal
- c) Ramp Signal
- d) Sinusoidal Signal
- e) Exponential Signal
- f) Signum Signal
- g) Rectangular Signal
- h) Sinc Signal



i) **Unit Impulse Signal:**

➤ A discrete-time (DT) unit impulse signal denoted as $\delta[n]$ is defined as

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

➤ Also known as unit sample sequence or unit impulse function.

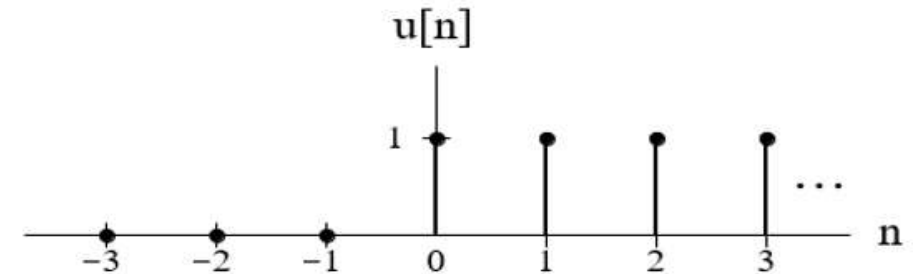
- Delayed impulse function can be expressed as

$$k \delta[n - k] = \begin{cases} k, & n = k \\ 0, & n \neq k \end{cases}$$

b) Unit Step Signal:

- The discrete-time unit step signal is defined as

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



- Delayed unit step signal can be expressed as

$$k u[n - k] = \begin{cases} k, & n \geq k \\ 0, & n < k \end{cases}$$

c) Ramp Function:

- The discrete-time unit ramp signal is defined as

$$r[n] = n u[n]$$

▪ **Relationships between impulse, step and ramp functions:**

- A discrete-time **unit step** is the **running sum** of the **unit impulse** as

$$u[n] = \sum_{m=-\infty}^n \delta[m]$$

or equivalently,

$$u[n] = \sum_{k=0}^{\infty} \delta[n - k]$$

and the discrete-time **unit impulse** is the **first difference** of the DT **unit step** , i.e.,

$$\delta[n] = u[n] - u[n - 1]$$

- Similarly, discrete-time **ramp signal** is the **running sum** of the **unit step signal** as

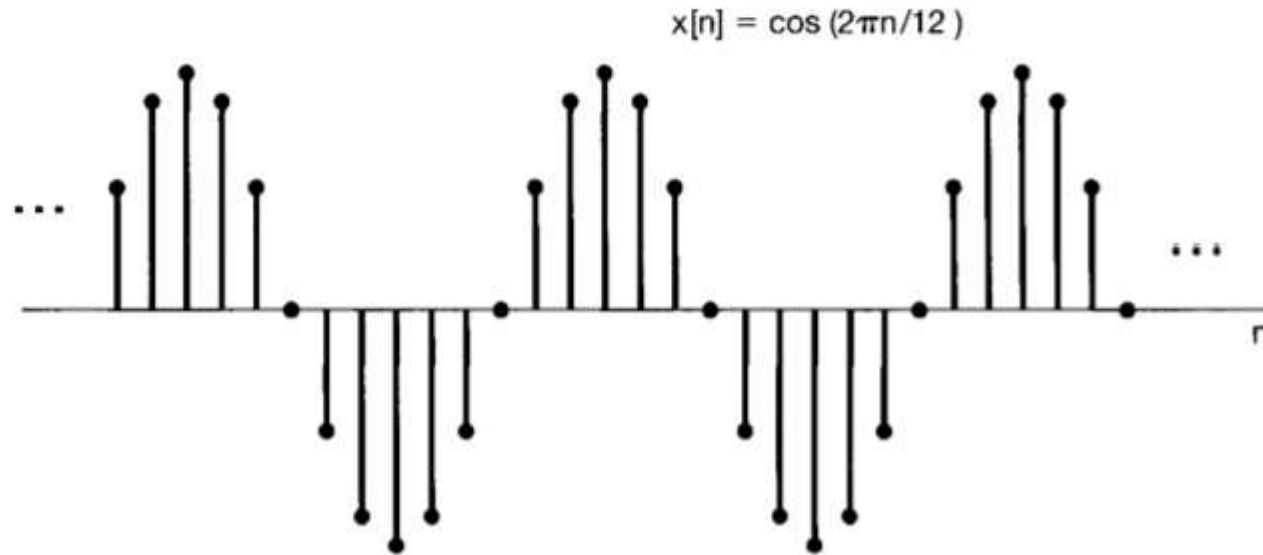
$$r[n] = \sum_{m=-\infty}^n u[m] \quad \text{(running sum of unit step)}$$

and the discrete-time **unit step** is the **first difference** of the DT **ramp signal** , i.e.,

$$u[n] = r[n] - r[n - 1] \quad \text{(first difference of unit ramp)}$$

d) Sinusoidal Signals:

- The signals varying with sine function of time or cosine function of time are called sinusoidal signals.



➤ The sinusoidal signal is of the form

$$x[n] = A \cos(\omega_0 n + \emptyset)$$

where, A = amplitude

ω_0 = frequency, and

$\emptyset$ = phase angle in radians

e) Exponential Signal:

- A discrete-time exponential signal is defined as

$$x[n] = c \alpha^n,$$

where, c and α are ,in general, complex numbers.

- The exponential nature of $x[n]$ is confirmed by defining

$$\alpha = e^\beta$$

then

$$x[n] = c e^{\beta n}$$

i. Real exponential:

- If c and α are real then $x[n]$ is real exponential signal. It exhibits one of the several types of behaviors as illustrated in fig.8 below:

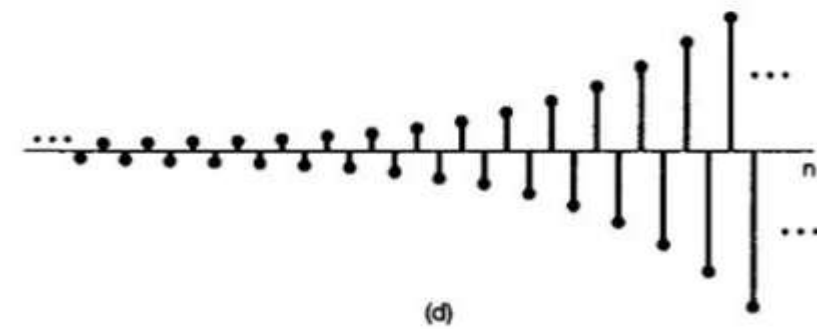
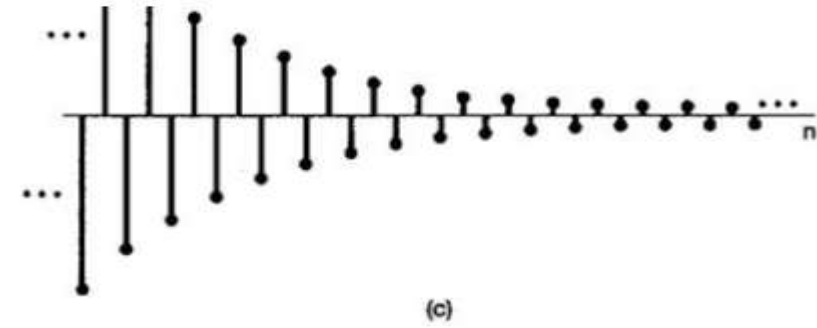
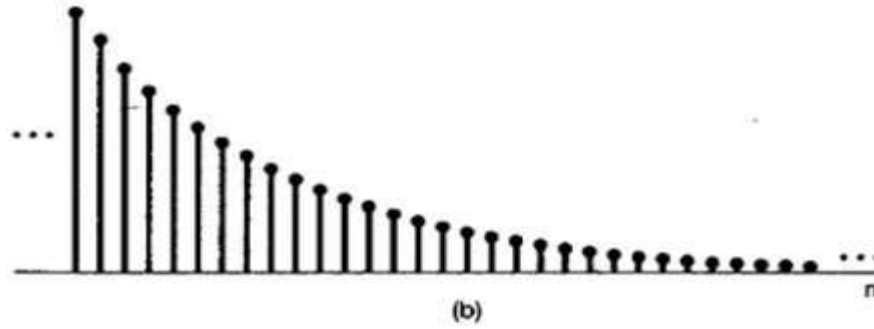
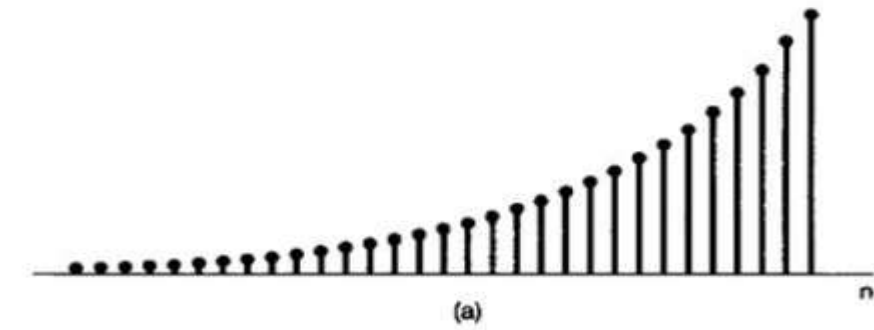


Fig: The real exponential signal

a) $\alpha > 1$

c) $-1 < \alpha < 0$

b) $0 < \alpha < 1$

d) $\alpha < -1$

ii. General Complex Exponential Signals:

➤ If c and α are expressed in polar form then the resulting exponential is

$$x[n] = c \alpha^n$$

$$x[n] = |c|e^{j\theta} |\alpha|^n e^{j\omega_0 n}$$

$$= |c||\alpha|^n e^{j(\omega_0 n + \theta)}$$

$$= |c||\alpha|^n \{\cos(\omega_0 n + \theta) + j\sin(\omega_0 n + \theta)\}$$

- Cases:

- i. For $|\alpha| = 1$, the real and imaginary parts of a complex exponential sequences are sinusoidal.
- ii. For $|\alpha| < 1$, they corresponds to sinusoidal sequences multiplied by a decaying exponential.
- iii. For $|\alpha| > 1$, they corresponds to sinusoidal sequences multiplied by a growing exponential.

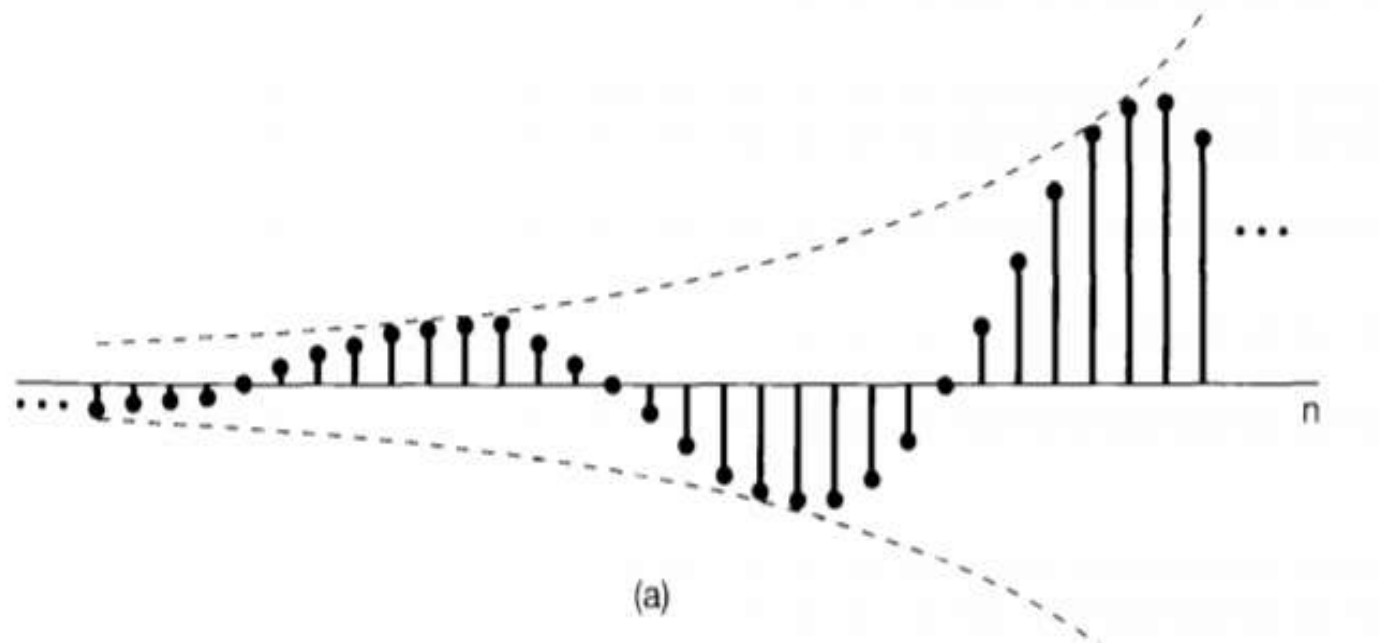
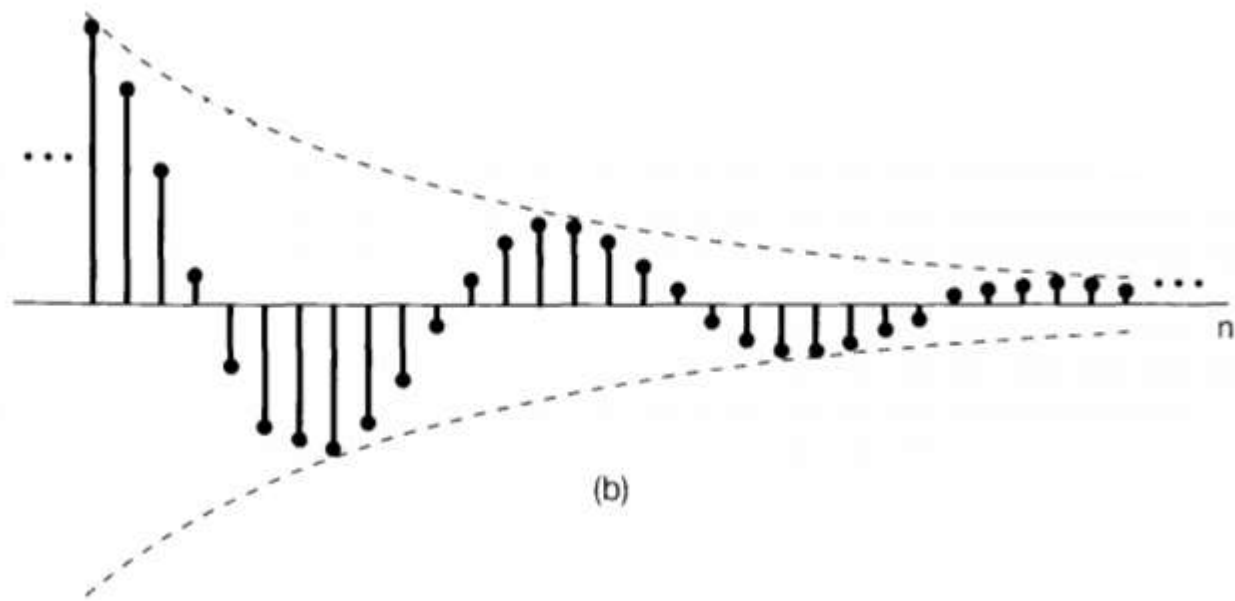


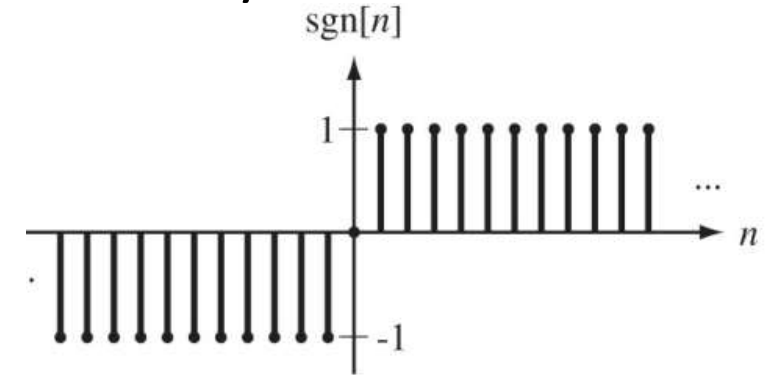
Fig: a) A growing DT sinusoidal signal
b) A decaying DT sinusoidal signal



f) Signum Signal:

➤ A discrete-time signum signal is denoted by $\text{sgn}[n]$ and defined by

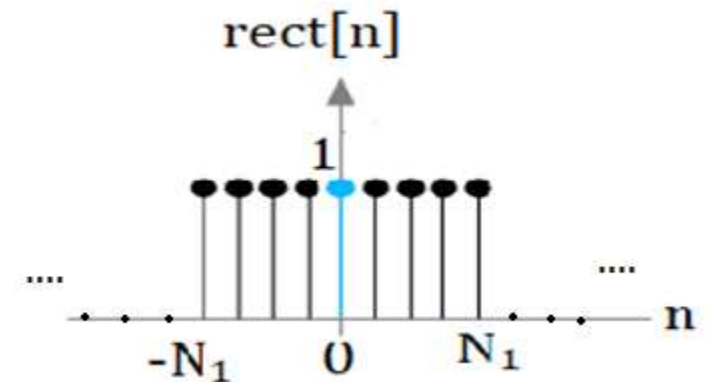
$$\text{sgn}[n] = \begin{cases} 1, & n > 0 \\ 0, & n = 0 \\ -1, & n < 0 \end{cases}$$



g) Rectangular Signal:

➤ It is defined as

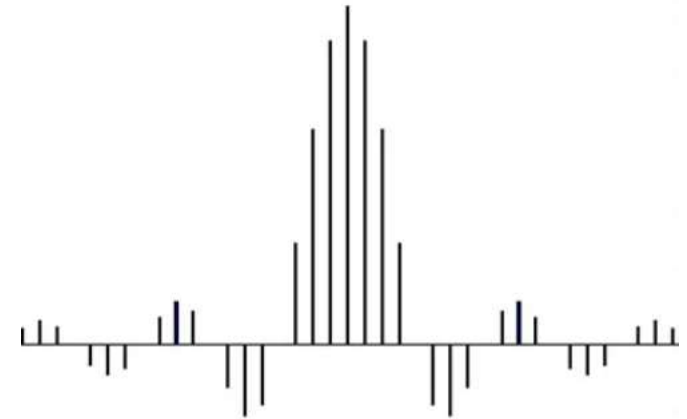
$$\text{rect}[n] = \begin{cases} 1, & -N_1 \leq n \leq N_1 \\ 0, & \text{Otherwise} \end{cases}$$



h) Sinc Signal:

- It is defined by

$$\text{Sinc}[an] = \begin{cases} 0, & an = \pm k\pi \\ \frac{\sin[an]}{an}, & \text{Otherwise} \end{cases}$$



□ Periodicity of Discrete-Time Signals:

- A discrete-time sinusoidal (or complex exponential) signal $x[n]$ is periodic if and only if its fundamental frequency ($\omega_0 = \frac{2\pi}{N}$) is rational.
- A discrete-time signal is periodic if $x[n + N] = x[n]$ for all n . Let a Sinusoidal signal is expressed as $x[n] = A\cos[\omega_0 n + \theta]$ where A =amplitude; ω_0 =fundamental frequency(radian/sec) ; θ = phase shift then
$$x[n + N] = A\cos\{\omega_0[n + N] + \theta\}$$
$$= A\cos\{[\omega_0 n + \theta] + \omega_0 N\}$$
- But, $\omega_0 N = 2\pi k$ then we can write

$$x[n + N] = A\cos[\omega_0 n + \theta]$$
$$= x[n]$$

- Therefore, a discrete-time signal is periodic when

$$\omega_0 N = 2\pi k \quad [k \text{ is integer}]$$

$$\omega_0 = \frac{2\pi k}{N}$$

- But

$$\omega_0 = 2\pi f_0$$

Therefore,

$$f_0 = \frac{k}{N} \text{ which is rational.}$$

- **Repeat the process for complex exponential signal.**

Example: Determine the fundamental frequency of the signal $x[n] = 1 + e^{j\frac{4\pi}{7}n} - e^{-j\frac{2\pi}{5}n}$.

Solution:

- Given DT signal is $x[n] = 1 + e^{j\frac{4\pi}{7}n} - e^{-j\frac{2\pi}{5}n}$

- Let $x[n] = x_1[n] + x_2[n] - x_3[n]$

- **Step-1:**

- Here , $x_1[n] = 1$
 $= e^{j2\pi n}$

- Compare with $x[n] = e^{j\omega_0 n}$, we have $\omega_0 = 2\pi$

➤ Then ,

$$N_1 = k \frac{2\pi}{\omega_0}$$

$N_1 = k \frac{2\pi}{2\pi} = 1 \times 1 = 1$ (because the fundamental period is the smallest possible integer. In this case which is 1)

➤ Similarly,

$$x_2[n] = e^{j\frac{4\pi}{7}n}$$

Then ,

$$\omega_0 = \frac{4\pi}{7} \quad \text{and} \quad N_2 = k \frac{2\pi}{\omega_0} = k \frac{2\pi}{4\pi/7} = 7$$

➤ Again

$$x_3[n] = e^{-j\frac{2\pi}{5}n}$$

Comparing with $x_3[n] = e^{-j\omega_0 n}$, we have $\omega_0 = \frac{2\pi}{5}$

Therefore,

$$N_3 = k \frac{2\pi}{\omega_0} = k \frac{2\pi}{\frac{2\pi}{5}} = 5$$

- **Step-2:**

- Since N_1, N_2 and N_3 are integers their ratios also rational.

i.e. $\frac{N_1}{N_2}$ and $\frac{N_2}{N_3}$ are also rational.

Hence the overall signal $x[n]$ is periodic.

- **Step-3:**

- The fundamental period (N) of the signal $x[n]$ is the LCM of N_1, N_2 and N_3

Therefore, $N = 1 \times 7 \times 5 = 35$

❑ Transformation of Independent Variable:

- Systems are used to process or manipulate signals. It involves a combinations of some basic operations
- There are two classes of operations as
 1. Operations performed on independent variables
 2. Operations performed on dependent variables

1. Operation performed on independent variable:

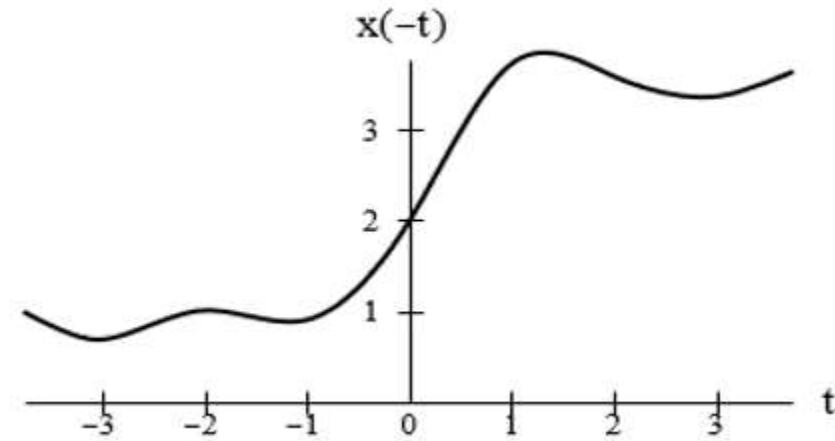
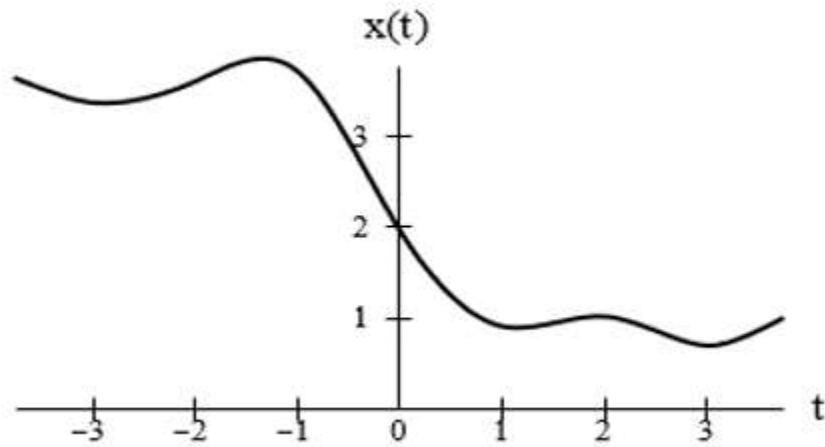
- Independent variable is time(DT or CT). There are 3 types of operations:

a) Time Reversal:

- It is also called the time folding or reflection. It is defined as

$$y[n] = x[-n] \text{ for DT signals.}$$

$$y(t) = x(-t) \text{ for CT signals.}$$



➤ Geometrically, the output signal is the reflection of the input signal about the vertical axis, at origin.

- Cases:

- i. Even signal is the same as its reflected version because we know

$$x(t) = x(-t) \text{ for all } t.$$

- ii. Odd signal is the negative of its reflected version because we know

$$x(-t) = -x(t) \text{ for all } t.$$

b) Time shifting:

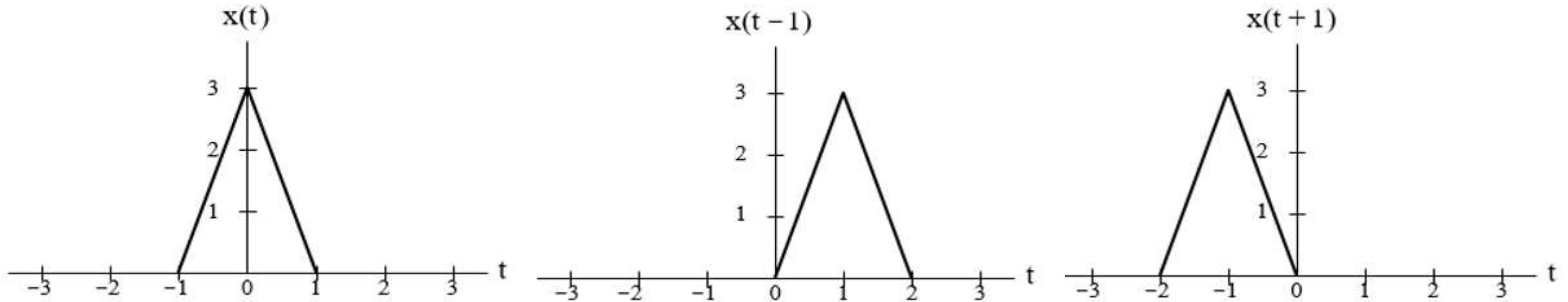
➤ Time shifting (also called translation) maps the input signal to the output signal given as

$$y(t) = x(t - t_0) \quad \text{for CT signals.}$$

$$y[n] = x[n - k] \quad \text{for DT signals.}$$

- Cases:

- If $t_0 > 0$ (or $k > 0$), the output signal is shifted towards right (positive shift).
- If $t_0 < 0$ (or $k < 0$), the output signal is shifted towards left (negative shift).



- $x(t - t_0)$: positive shift
- $x(t + t_0)$: negative shift

c) Time Scaling:

➤ Time scaling maps the input signal to the output signal as given by

$$y(t) = x(at), \quad a \text{ is strictly a positive real number (for CT signals).}$$

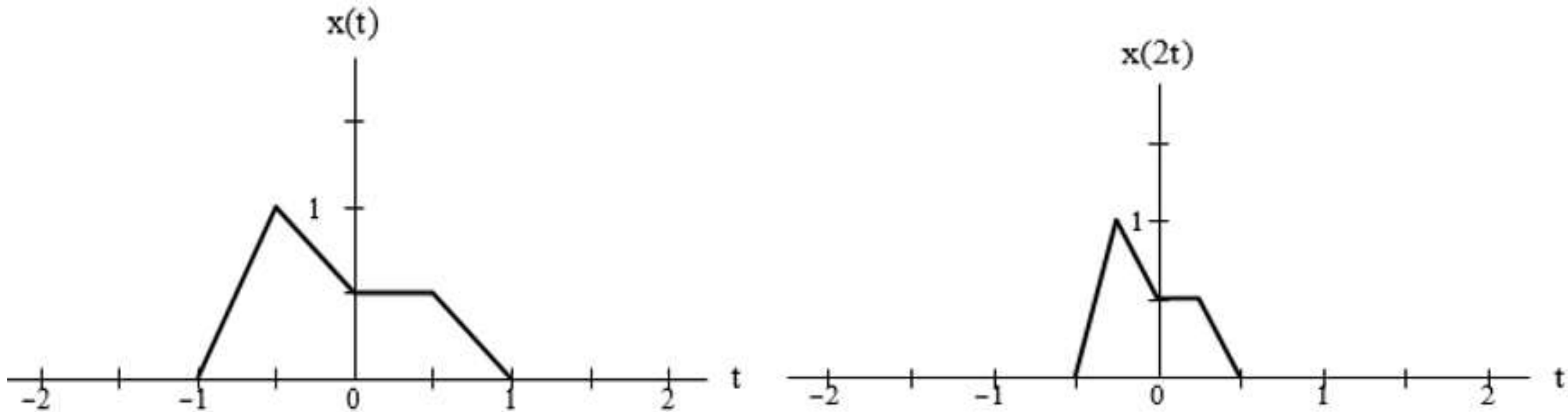
$$y[n] = x[kn], \quad (\text{for DT signals})$$

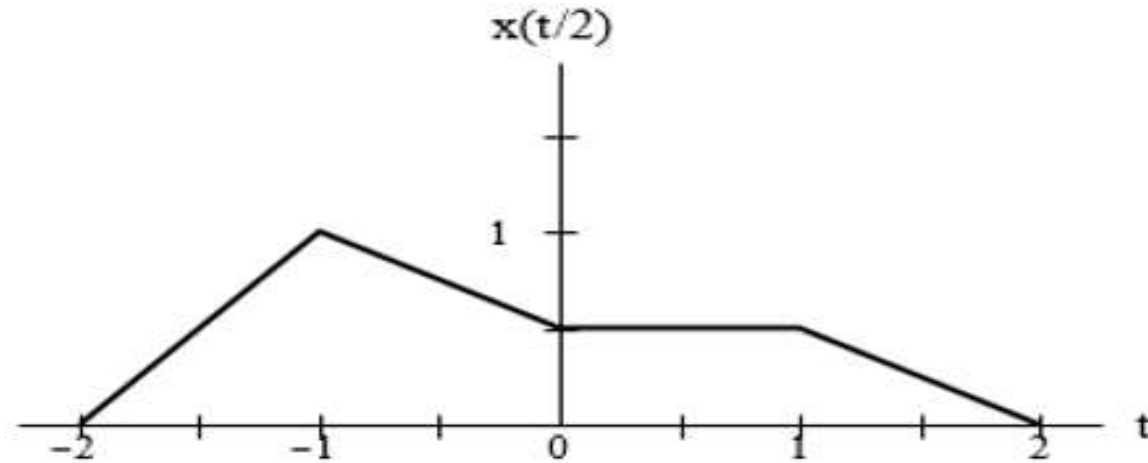
■ Cases:

i. If $a > 1$ (or $k > 1$), the output signal is compressed version.

ii. If $a < 1$ (or $k < 1$), the output signal is expanded version.

➤ Note that if $k > 1$, some of the values (non integer values) are lost.





❑ Discrete-Time Fourier Series (DTFS) and Properties:

1. Introduction:

- **Fourier analysis** involves the resolution of certain classes of signals into the superposition of sines and cosines (sinusoidal components) or, equivalently complex exponential signals.
- The output of a system to a sinewave is another sinewave of the same frequency (but with a different phase and amplitude) under two conditions:
 - a) If the system is linear, and
 - b) If the system is time-invariant.

- **Periodic signals** have Fourier series representation(Continuous-Time Fourier Series(CTFS) and Discrete-Time Fourier Series(DTFS))
- **Aperiodic (or non-periodic)** signals have Fourier transform representation (Continuous-Time Fourier Transform(CTFT) and Discrete-Time Fourier Transform (DTFT)).

2. Fourier Series Definition:

- A general periodic signal or function can be represented as an infinite sum of sine and cosine function (or, equivalently complex exponential function) which are then itself periodic. This series of sine and cosine terms (or, complex exponential terms) is called **Fourier series**.
- **Fourier series** can be studied as under:
 - 1) Continuous-Time Fourier Series(CTFS) , and
 - 2) Discrete-Time Fourier Series(DTFS)

3. Discrete-Time Fourier Series(DTFS) Representation:

- DTFS is a finite series and there is no issue of convergence.

a. Linear Combinations of Harmonically related Complex Exponentials:

- A DT periodic signal is defined as

$$x[n] = x[n + N] \text{ for all } n. \quad \text{.....1}$$

The smallest value of N that satisfy the eq.1 is called fundamental period.

Then the fundamental frequency is $\omega_0 = \frac{2\pi}{N}$.

- The periodic complex exponential is

$$x[n] = e^{j\omega_0 n} \quad \text{.....2}$$

- Harmonically related complex exponential is

$$S_k[n] = e^{jk\omega_0 n} \quad \text{.....3}$$

where,

$$N = \frac{2\pi}{|k|\omega_0} \text{ and } \omega_0 = \frac{2\pi}{|k|N}$$

- Then the linear combinations of harmonically related complex exponentials is

$$x[n] = \sum_k D_k S_k[n] = \sum_k D_k e^{jk\omega_0 n} = \sum_k D_k e^{jk\frac{2\pi}{N}n}$$

- Since , $S_k[n]$ are distinct only over a range of N successive values of k , we have

$$x[n] = \sum_{k=N} D_k S_k[n] = \sum_{k=N} D_k e^{jk\omega_0 n} = \sum_{k=N} D_k e^{jk\frac{2\pi}{N}n} \dots\dots\dots 4$$

- Eq.4 is referred to as the **DTFS** and D_k as the **Fourier series coefficients(or spectral coefficients)**.

b. Determination of DTFS Representation:

- Let DT periodic signal $x[n]$ of period N is

$$x[n + N] = x[n] \quad \text{for all } N. \dots\dots\dots 5$$

- The Fourier series representation for $x[n]$ is

$$x[n] = \sum_{k=N} D_k e^{jk\omega_0 n} = \sum_{k=N} D_k e^{jk\frac{2\pi}{N}n} \dots\dots\dots 6$$

where D_k are the Fourier series coefficients.

- Multiplying eq.2 on both sides by $e^{-jl\frac{2\pi}{N}n}$ and summing over N terms, we get

$$\sum_{n=N} x[n] e^{-jl\frac{2\pi}{N}n} = \sum_{n=N} \sum_{k=N} D_k e^{j(k-l)\frac{2\pi}{N}n}$$

$$\sum_{n=N} x[n] e^{-jl\frac{2\pi}{N}n} = \sum_{k=N} D_k \left\{ \sum_{n=N} e^{j(k-l)\frac{2\pi}{N}n} \right\} \dots\dots\dots 7$$

- But we know that

$$\sum_{n=N} \left(e^{j(k-l)\frac{2\pi}{N}n} \right)_n = \begin{cases} N, & k - l = 0, \pm N, \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases}$$

- Then RHS of eq.3 reduces to

$$\sum_{n=N} x[n] e^{-jl\frac{2\pi}{N}n} = D_l N$$

$$D_l = \frac{1}{N} \sum_{n=N} x[n] e^{-jl\frac{2\pi}{N}n}$$

- In general,

$$D_k = \frac{1}{N} \sum_{n=-N}^N x[n] e^{-jk\frac{2\pi}{N}n} \dots\dots\dots 8$$

This is the expression for Fourier series coefficients.

- Eq.4 is also called **analysis equation** and equation 2 is called **synthesis equation**.

- **Example: Show that the Fourier series coefficients are periodic.**

Proof:

- We know D_k provides the frequency domain description of $x[n]$. The D_k represents the amplitude and phase associated with the frequency component $e^{jk\frac{2\pi}{N}n}$.

- Since $e^{jk\frac{2\pi}{N}n}$ are periodic with period N , D_k are also periodic.

$$D_{K+N} = \frac{1}{N} \sum_{n=-N}^N x[n] e^{-j(k+N)\frac{2\pi}{N}n}$$

$$D_{K+N} = \frac{1}{N} \sum_{n=-N}^N x[n] e^{-jk\frac{2\pi}{N}n} e^{-j2\pi n}$$

$$D_{K+N} = \frac{1}{N} \sum_{n=N} x[n] e^{-jk \frac{2\pi}{N} n}$$

$$D_{K+N} = D_k$$

Example: Find the Fourier series coefficients of the signal

$$x[n] = 1 + \sin \frac{2\pi}{N} n + 3 \cos \left(\frac{2\pi}{N} n \right) + \cos \left(\frac{4\pi}{N} n + \frac{\pi}{2} \right)$$

Solution:

➤ Given, $x[n] = 1 + \sin \frac{2\pi}{N} n + 3 \cos \left(\frac{2\pi}{N} n \right) + \cos \left(\frac{4\pi}{N} n + \frac{\pi}{2} \right)$

$$= 1 + \left\{ \frac{e^{j \frac{2\pi}{N} n} - e^{-j \frac{2\pi}{N} n}}{2j} \right\} + 3 \left\{ \frac{e^{j \frac{2\pi}{N} n} + e^{-j \frac{2\pi}{N} n}}{2} \right\} + \left\{ \frac{e^{j \left(\frac{4\pi}{N} n + \frac{\pi}{2} \right)} + e^{-j \left(\frac{4\pi}{N} n + \frac{\pi}{2} \right)}}{2} \right\}$$

$$= \left(\frac{1}{2} e^{-j \frac{\pi}{2}} \right) e^{-j 2 \frac{2\pi}{N} n} + \left(\frac{3}{2} - \frac{1}{2j} \right) e^{-j \frac{2\pi}{N} n} + 1 + \left(\frac{3}{2} + \frac{1}{2j} \right) e^{j \frac{2\pi}{N} n} +$$

$$\left(\frac{1}{2} e^{j \frac{\pi}{2}} \right) e^{j 2 \frac{2\pi}{N} n} \dots \dots \dots (i)$$

➤ Comparing eq.i with the equation $x[n] = \sum_{k=-N}^N D_k e^{jk\frac{2\pi}{N}n}$
i.e., $x[n] = \dots + D_{-2}e^{-j2\frac{2\pi}{N}n} + D_{-1}e^{-j\frac{2\pi}{N}n} + D_0 + D_1e^{j\frac{2\pi}{N}n} + D_2e^{j2\frac{2\pi}{N}n} + \dots$

we get, $D_0 = 1$, $D_1 = \frac{3}{2} + \frac{1}{2j} = \frac{3}{2} - \frac{j}{2}$, $D_{-1} = \frac{3}{2} - \frac{1}{2j} = \frac{3}{2} + \frac{j}{2}$,

$$D_2 = \frac{1}{2}e^{j\frac{\pi}{2}} = j\frac{1}{2}$$

$$D_{-2} = \frac{1}{2}e^{-j\frac{\pi}{2}} = -j\frac{1}{2}$$

Example: Determine the spectrum of the following signals.

a) $x[n] = \cos \frac{\pi}{3}n$

b) $x[n] = \{1,1,0,0\}$

Solution:

a) Given, $x[n] = \cos \frac{\pi}{3}n$

we know $x[n] = \cos \omega_0 n$, we get $\omega_0 = \frac{\pi}{3}$ and $N = \frac{2\pi}{\omega_0} = 6$

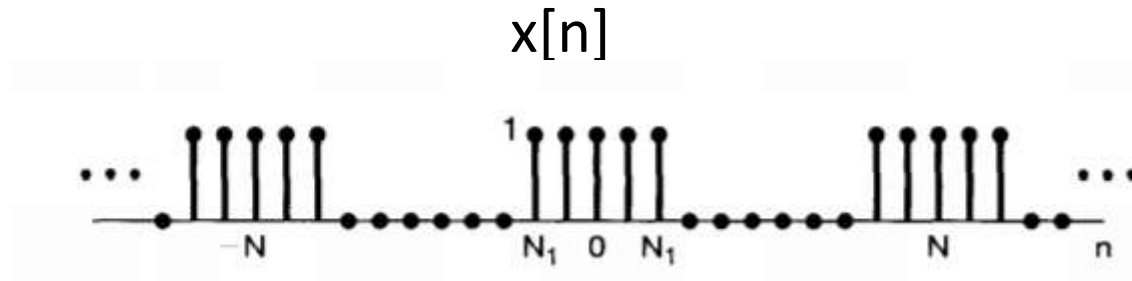
Now, $\cos \frac{\pi}{3}n = \frac{1}{2}(e^{j\frac{\pi}{3}n} + e^{-j\frac{\pi}{3}n})$

$$\begin{aligned}
\cos \frac{\pi}{3} n &= \frac{1}{2} e^{j\frac{2\pi}{6}n} + \frac{1}{2} e^{-j\frac{2\pi}{6}n} \\
&= \frac{1}{2} e^{j\frac{2\pi}{6}n} + \frac{1}{2} e^{j\frac{2\pi(5-6)}{6}n} \\
&= \frac{1}{2} e^{j\frac{2\pi}{6}n} + \frac{1}{2} e^{j\frac{5(2\pi)}{6}n} \\
&= D_1 + D_5 \quad ; D_0, D_2, D_3, D_4 = 0
\end{aligned}$$

And

$$D_k = \frac{1}{6} \sum_{n=0}^5 x[n] e^{-jk\frac{2\pi}{6}n}$$

Example: Determine the Fourier series coefficients of the periodic signal shown in figure below



Solution:

➤ The given signal is the periodic square wave. The Fourier series coefficient is given by

$$D_k = \frac{1}{N} \sum_{n=-N}^N x[n] e^{-jk\frac{2\pi}{N}n} ; \omega_0 = \frac{2\pi}{N}$$

$$= \frac{1}{N} \sum_{n=-N_1}^{N_1} 1 e^{-jk\omega_0 n} \dots\dots\dots 1$$

➤ Let, $m = n + N_1$ in eq.1 then

$$D_k = \frac{1}{N} \sum_{m=0}^{2N_1} 1 e^{-jk\omega_0(m-N_1)} = \frac{1}{N} e^{jk\omega_0 N_1} \sum_{m=0}^{2N_1} (e^{-jk\omega_0})^m$$

$$\blacktriangleright \text{ Or } D_k = \frac{1}{N} e^{jk\omega_0 N_1} \left\{ \frac{1 - e^{-jk\omega_0(2N_1+1)}}{1 - e^{-jk\omega_0}} \right\} = \frac{1}{N} \left\{ \frac{e^{jk\omega_0 N_1} - e^{-jk\omega_0 2N_1} e^{-jk\omega_0} e^{jk\omega_0 N_1}}{1 - e^{-jk\omega_0}} \right\}$$

$$= \frac{1}{N} \left\{ \frac{e^{jk\omega_0 N_1} - e^{-jk\omega_0 N_1} e^{jk\omega_0}}{1 - e^{-jk\omega_0}} \right\}$$

$$= \frac{1}{N} \left\{ \frac{e^{-jk\frac{\omega_0}{2}} \{e^{jk\frac{\omega_0}{2}} e^{jk\omega_0 N_1} - e^{-jk\omega_0 N_1} e^{-jk\frac{\omega_0}{2}}\}}{e^{-jk\frac{\omega_0}{2}} \{e^{jk\frac{\omega_0}{2}} - e^{-jk\frac{\omega_0}{2}}\}} \right\}$$

$$= \frac{1}{N} \left[\frac{\{e^{jk\omega_0(N_1+\frac{1}{2})} - e^{-jk\omega_0(N_1+\frac{1}{2})}\} / 2j}{\{e^{jk\frac{\omega_0}{2}} - e^{-jk\frac{\omega_0}{2}}\} / 2j} \right]$$

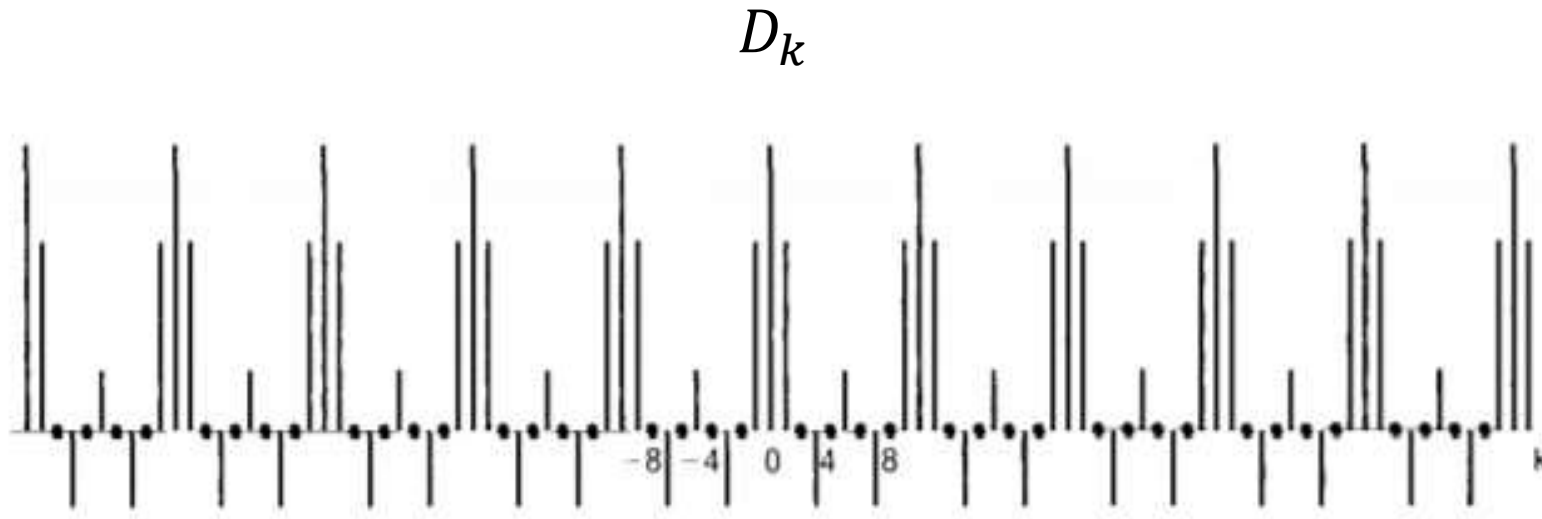
$$D_k = \frac{1}{N} \left[\frac{\sin\{k\omega_0(N_1+\frac{1}{2})\}}{\sin(k\frac{\omega_0}{2})} \right] = \frac{1}{N} \left[\frac{\sin\left\{\frac{2\pi k(N_1+\frac{1}{2})}{N}\right\}}{\sin\left(\frac{\pi k}{N}\right)} \right], \quad k \neq 0, \pm N, \pm 2N, \dots \dots \dots$$

and
$$D_k = \frac{2N_1+1}{N}, \quad k = 0, \pm N, \pm 2N, \dots$$

- When $k = 0$, $D_k = \frac{1}{N} e^{jk\omega_0 N_1} \sum_{m=0}^{2N_1} (e^{-jk\omega_0})^m = \frac{1}{N} \sum_{m=0}^{2N_1} (1)^m = \frac{2N_1+1}{N}$

-

➤ The coefficients D_k for $2N_1 + 1 = 5$ is sketched for $N = 10$ is shown below



c. Properties of DTFS:

- Properties are used to develop conceptual fundamentals and in reducing the complexity of Fourier series of many signals.

1) Linearity:

➤ Let

$$\begin{array}{ccc} x[n] & \xleftrightarrow{FS} & D_k \\ y[n] & \xleftrightarrow{FS} & Y_k \end{array}$$

and both are periodic with period N . Then the linear combination will be the periodic with period N .

i.e.,
$$z[n] = ax[n] + by[n] \xleftrightarrow{FS} Z_k = aD_k + bY_k.$$

Proof:

➤ We know,

$$\begin{aligned} Z_k &= \frac{1}{N} \sum_{n=N} z[n] e^{-jk\frac{2\pi}{N}n} \\ &= \frac{1}{N} \sum_{n=N} \{ax[n] + by[n]\} e^{-jk\frac{2\pi}{N}n} \\ &= a \left\{ \frac{1}{N} \sum_{n=N} x[n] e^{-jk\frac{2\pi}{N}n} \right\} + b \left\{ \frac{1}{N} \sum_{n=N} y[n] e^{-jk\frac{2\pi}{N}n} \right\} \end{aligned}$$

Khemraj koirala, WRC

➤ Therefore,

$$Z_k = aD_k + bY_k$$

2) Time Shifting:

➤ Let

$$x[n] \xleftrightarrow{FS} D_k$$

then

$$y[n] = x[n - n_0] \xleftrightarrow{FS} Y_k = D_k e^{-jk \frac{2\pi}{N} n_0}$$

Proof:

➤ We have,

$$\begin{aligned} Y_k &= \frac{1}{N} \sum_{n=N} y[n] e^{-jk \frac{2\pi}{N} n} \\ &= \frac{1}{N} \sum_{n=N} \{x[n - n_0]\} e^{-jk \frac{2\pi}{N} n} \end{aligned}$$

➤ Let $n - n_0 = s$ and s will range over an interval of duration N . Then

$$\begin{aligned} Y_k &= \frac{1}{N} \sum_{s=N} x[s] e^{-jk \frac{2\pi}{N} (s+n_0)} \\ Y_k &= \left\{ \frac{1}{N} \sum_{s=N} x[s] e^{-jk \frac{2\pi}{N} s} \right\} e^{-jk \frac{2\pi}{N} n_0} \end{aligned}$$

➤ Therefore,

$$Y_k = D_k e^{-jk \frac{2\pi}{N} n_0} \quad \text{where } D_k = \frac{1}{N} \sum_{s=N} x[s] e^{-jk \frac{2\pi}{N} s}$$

- Since $|Y_k| = |D_k|$, time shifting property does not change the amplitude. It only changes in the phase spectrum.

3) Frequency Shifting:

- If $x[n] \xleftrightarrow{FS} D_k$
then $x[n]e^{jm\frac{2\pi}{N}n} \xleftrightarrow{FS} D_{k-m}$

Proof:

- We know, $D_k = \frac{1}{N} \sum_{n=N} x[n] e^{-jk\frac{2\pi}{N}n}$

This implies that $\frac{1}{N} \sum_{n=N} \{x[n]e^{jm\frac{2\pi}{N}n}\} e^{-jk\frac{2\pi}{N}n} = \frac{1}{N} \sum_{n=N} x[n]e^{-j(k-m)\frac{2\pi}{N}n} = D_{k-m}$

- Therefore, $x[n]e^{jm\frac{2\pi}{N}n} \xleftrightarrow{FS} D_{k-m}$

4) Time Reversal:

➤ Let $x[n]$ $\xleftrightarrow{FS} D_k$
 then $x[-n]$ $\xleftrightarrow{FS} D_{-k}$

Proof:

➤ We know, $x[n] = \sum_{k=-N}^N D_k e^{jk\frac{2\pi}{N}n}$ 1

➤ Put, $n = -n$, we get

$$x[-n] = \sum_{k=-N}^N D_k e^{-jk\frac{2\pi}{N}n}$$

➤ Let $k = -m$ in RHS , we get

$$x[-n] = \sum_{m=-N}^N D_{-m} e^{jm\frac{2\pi}{N}n}, \text{ (Period remains unchanged and positive)}$$

➤ In general, $x[-n] = \sum_{k=-N}^N D_{-k} e^{jk\frac{2\pi}{N}n}$ 2

➤ From eqs. 1 and 2 we have

$$x[-n] \xleftrightarrow{FS} D_{-k}$$

- When a periodic signal undergoes time reversal, the period N of the signal remains unchanged. It results in a time reversal of the corresponding sequence of the Fourier series coefficients.

- Cases:

I. When $x[n]$ is even then

$$D_k = D_{-k}$$

II. When $x[n]$ is odd then

$$D_{-k} = -D_k$$

5) Conjugation and Conjugate Symmetry:

- Taking the complex conjugate of a DT signal $x[n]$ has the effect of complex conjugation and time reversal on the corresponding Fourier series coefficients.

➤ If

$$x[n] \xleftrightarrow{FS} D_k$$

then

$$x^*[n] \xleftrightarrow{FS} D_{-k}^*$$

Proof:

➤ We have,

$$x[n] = \sum_{k=-N}^N D_k e^{jk \frac{2\pi}{N} n} \dots\dots\dots 1$$

➤ Taking complex conjugate on both sides of eq.1, we get

$$\{x[n]\}^* = \{\sum_{k=-N} D_k e^{jk\frac{2\pi}{N}n}\}^*$$

$$x^*[n] = \sum_{k=-N} D_k^* e^{-jk\frac{2\pi}{N}n}$$

➤ Put $k = -k$, we get

$$x^*[n] = \sum_{k=-N} D_{-k}^* e^{jk\frac{2\pi}{N}n} \dots\dots\dots 2$$

which shows that

$$x^*[n] \quad \xleftrightarrow{FS} \quad D_{-k}^*$$

6) Multiplication:

➤ If $x[n] \xleftrightarrow{FS} D_k$

and $y[n] \xleftrightarrow{FS} Y_k$

then $z[n] = x[n]y[n] \xleftrightarrow{FS} Z_k = \sum_{l=N} D_l Y_{k-l}$

provided that $x[n]$, $y[n]$ and the product $x[n]y[n]$, also periodic with period N .

Proof:

➤ Given , $x[n] = \sum_{k=N} D_k e^{jk\frac{2\pi}{N}n}$

and $y[n] = \sum_{k=N} Y_k e^{jk\frac{2\pi}{N}n}$

➤ We know, $Z_k = \frac{1}{N} \sum_{n=N} \{z[n]\} e^{-jk\frac{2\pi}{N}n}$

$$Z_k = \frac{1}{N} \sum_{n=N} \{x[n]y[n]\} e^{-jk\frac{2\pi}{N}n}$$

$$Z_k = \frac{1}{N} \sum_{n=N} \{ \sum_{l=N} D_l e^{jl \frac{2\pi}{N} n} \} y[n] e^{-jk \frac{2\pi}{N} n}$$

$$Z_k = \sum_{l=N} D_l \{ \frac{1}{N} \sum_{n=N} y[n] e^{-j(k-l) \frac{2\pi}{N} n} \}$$

$$Z_k = \sum_{l=N} D_l Y_{k-l}$$

➤ Therefore,

$$z[n] = x[n]y[n] \xleftrightarrow{FS} Z_k = \sum_{l=N} D_l Y_{k-l}$$

7) Convolution:

➤ If

$$x[n] \xleftrightarrow{FS} D_k$$

$$z[n] \xleftrightarrow{FS} Z_k$$

and

$$y[n] \xleftrightarrow{FS} Y_k$$

then

$$y[n] = x[n] * z[n] \xleftrightarrow{FS} Y_k = N D_k Z_k$$

where $x[n]$, $y[n]$, $z[n]$ are periodic with period N . And $y[n]$ is the convolution of $x[n]$ and $z[n]$.

Proof:

- The convolution of periodic signal is defined as

$$y[n] = \sum_{s=-N}^N x[s]z[n-s]$$

- From time-shifting property, we have

$$z[n-s] \xleftrightarrow{FS} Z_s e^{-jk\omega_0 s} \quad \omega_0 = \frac{2\pi}{N}$$

$$\begin{aligned} \text{Thus, } \sum_{s=-N}^N x[s]z[n-s] &\xleftrightarrow{FS} \sum_{s=-N}^N x[s]Z_s e^{-jk\omega_0 s} = N \left\{ \frac{1}{N} \sum_{s=-N}^N x[s] e^{-jk\omega_0 s} \right\} Z_s \\ &= ND_s Z_s \\ &= Y_s \end{aligned}$$

- Since s is dummy variable, we can write

$$y[n] = x[n] * z[n] \xleftrightarrow{FS} Y_k = ND_k Z_k$$

8) Parseval's Relation:

- It states that the total average power of a periodic signal $x[n]$ is equal to the sum of the average power of its harmonic components (or Fourier components).

Mathematically

$$P_{av} = \frac{1}{N} \sum_{n=-N}^N |x[n]|^2 = \sum_{k=-N}^N |D_k|^2$$

Proof:

➤ We know, $P_{av} = \frac{1}{N} \sum_{n=-N}^N |x[n]|^2$

$$= \frac{1}{N} \sum_{n=-N}^N x[n] x^*[n]$$

$$= \frac{1}{N} \sum_{n=-N}^N x[n] \left\{ \sum_{k=-N}^N D_k^* e^{-jk\omega_0 n} \right\}$$

$$= \sum_{k=-N}^N D_k^* \left\{ \frac{1}{N} \sum_{n=-N}^N x[n] e^{-jk\omega_0 n} \right\}$$

$$= \sum_{k=-N}^N D_k D_k^*$$

$$= \sum_{k=-N}^N |D_k|^2$$

- $|D_k|^2$ is the average power in the k^{th} harmonic component (or k^{th} Fourier series component). Since $|D_k|^2$ is not a negative quantity, the power spectrum is symmetric spectrum.

❑ Discrete-Time Fourier Transform (DTFT) Representation:

- In periodic signals, complex exponentials are harmonically related.
- For aperiodic or energy signals, they are infinitesimally close in frequency and the resulting spectrum of coefficients is called the **Fourier transform**.
- An aperiodic signal can be viewed as periodic signal with infinite period($N_0 (T_0) \rightarrow \infty$). As $N_0 (T_0) \rightarrow \infty$, $\omega_0 = \frac{2\pi}{N_0} (\omega_0 = \frac{2\pi}{T_0}) \rightarrow 0$ (i.e., $d\omega$) and harmonically related complex exponential become closure in frequency ω_0 .
- The Fourier transform is studied as under:
 - 1) Continuous-Time Fourier Transform(CTFT)
 - 2) Discrete-Time Fourier Transform(DTFT)

1. Discrete-Time Fourier Transform (DTFT) Representation:

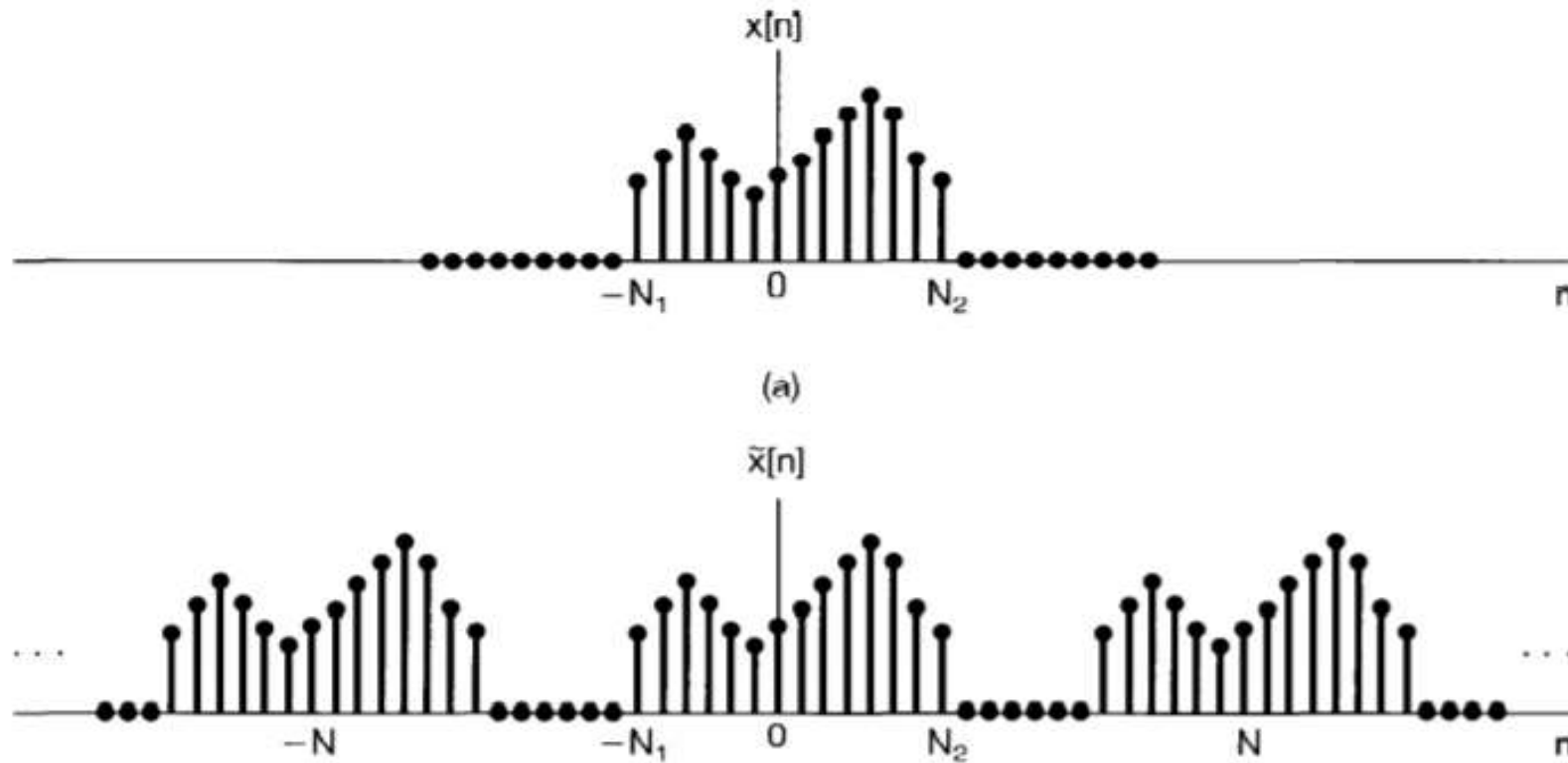


Fig.1:a) Aperiodic signal $x[n]$, (having finite duration)
b) Construction of periodic signal $\tilde{x}[n]$ by periodic extension of $x[n]$

➤ Fig (a) shows an aperiodic signal $x[n]$. Fig (b) shows a periodic signal $\tilde{x}[n]$ formed by periodically repeating the signal $x[n]$. The period N is made large enough to avoid overlap between the repeating pulses.

➤ From fig (a) and (b), we can write

$$\lim_{N \rightarrow \infty} \tilde{x}[n] = x[n] \text{ for any finite value of } n. \quad \text{.....1}$$

➤ Now, from DTFS, we can write

$$\tilde{x}[n] = \sum_{k=-\frac{N}{2}}^{\frac{N}{2}} D_k e^{jk\omega_0 n}, \text{ where } \omega_0 = \frac{2\pi}{N} \text{ is the fundamental}$$

frequency, and $N = \frac{2\pi}{\omega_0}$ is the fundamental period.2

and

$$D_k = \frac{1}{N} \sum_{n=-\frac{N}{2}}^{\frac{N}{2}} \tilde{x}[n] e^{-jk\omega_0 n} \quad \text{.....3}$$

➤ Since $\tilde{x}[n] = x[n]$ over a period that includes the interval such that $-N_1 \leq n \leq N_2$.

Then

$$D_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} x[n] e^{-jk\omega_0 n}$$

Or

$$D_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} x[n] e^{-jk\frac{2\pi}{N}n}$$

$$D_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\omega_0 n} \quad \text{.....4}$$

in eq.4 $x[n] = 0$ outside the interval $-N_1 \leq n \leq N_2$.

➤ Let us define, $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \quad \text{.....5}$

then from eq.4 and 5, we have

$$D_k = \frac{1}{N} X(e^{jk\omega_0}) \quad \text{.....6}$$

➤ From eq.2 and eq.6, we have

$$\begin{aligned} \tilde{x}[n] &= \sum_{k=-\frac{N}{2}}^{\frac{N}{2}} \left\{ \frac{1}{N} X(e^{jk\omega_0}) \right\} e^{jk\omega_0 n} \\ \tilde{x}[n] &= \frac{1}{2\pi} \sum_{k=-\frac{N}{2}}^{\frac{N}{2}} X(e^{jk\omega_0}) e^{jk\omega_0 n} \omega_0 \quad \text{.....7} \end{aligned}$$

➤ Since, $\lim_{N \rightarrow \infty} \tilde{x}[n] = x[n]$, we can write

$$\lim_{N \rightarrow \infty} \omega_0 = \lim_{N \rightarrow \infty} \frac{2\pi}{N} \rightarrow d\omega \text{ (infinitesimal small) and summation}$$

becomes integration.

also, $\lim_{\substack{N \rightarrow \infty \\ k \rightarrow \infty}} k \omega_0 = \lim_{\substack{N \rightarrow \infty \\ k \rightarrow \infty}} k \frac{2\pi}{N} = \omega$

then eq.7 can be written as

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \quad \text{.....8}$$

eq.8 can also be written as

$$x[n] = \frac{1}{2\pi} \int X(e^{j\omega}) e^{j\omega n} d\omega$$

➤ Eq.8 is the **synthesis equation or inverse(reverse) Fourier transform** and eq.5 is called **analysis equation or direct Fourier transform**.

- Note that the total interval of integration is 2π because eq.5 is 2π periodic.

$$\begin{aligned} X(e^{j(\omega+2\pi k)}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega+2\pi k)n} \\ &= \left\{ \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right\} e^{-j2\pi kn} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \quad (\text{since } e^{-j2\pi kn} = 1) \\ &= X(e^{j\omega}) \end{aligned}$$

▪ Convergence(or existence) of DTFT:

➤ The DTFT is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

➤ Because $|e^{-j\omega n}| = 1$, from above equation, $X(e^{j\omega})$ is converged if $x[n]$ is absolutely summable. That is,

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

➤ This condition is sufficient but necessary for the existence of $X(e^{j\omega})$.

• Examples:

1) Determine the DTFT of the following signals:

a) $x[n] = a^n u[n], |a| < 1$

b) $x[n] = a^{|n|}, |a| < 1$

c) $x[n] = \begin{cases} 1, & |n| \leq N_1 \\ 0, & |n| > N_1 \end{cases}$

d) $x[n] = a^n u[-n - 1]$

a) Solution:

➤ We know the DTFT is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

Where

$$x[n] = a^n u[n], |a| < 1$$

➤ Then

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} a^n u[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (ae^{-j\omega})^n$$

➤ Therefore , $X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$ provided that $|ae^{-j\omega}| < 1$

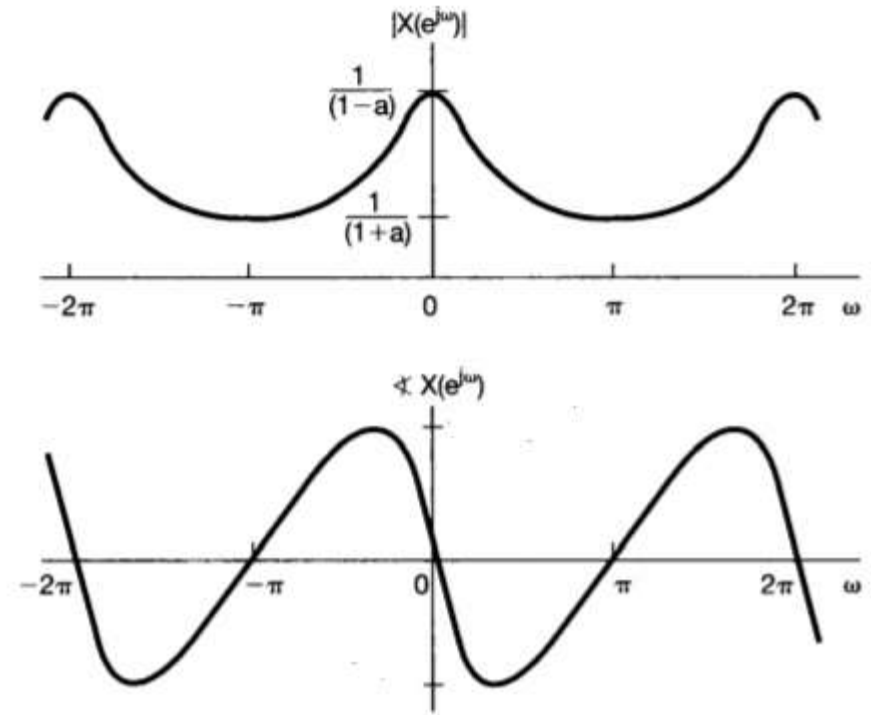
➤ The magnitude and phase of $X(e^{j\omega})$ are:

$$\bullet \quad |X(e^{j\omega})| = \frac{1}{1-a(\cos\omega - j\sin\omega)} = \frac{1}{(1-a\cos\omega) + j\sin\omega} = \frac{1}{\sqrt{(1-a\cos\omega)^2 + (a\sin\omega)^2}}$$

$$|X(e^{j\omega})| = \frac{1}{\sqrt{1+a^2-2a\cos\omega}}$$

$$\bullet \quad \theta(j\omega) = -\tan^{-1}\left[\frac{a\sin\omega}{1-a\cos\omega}\right]$$

➤ The plots for $a > 0$ is shown in fig.



b) Solution:

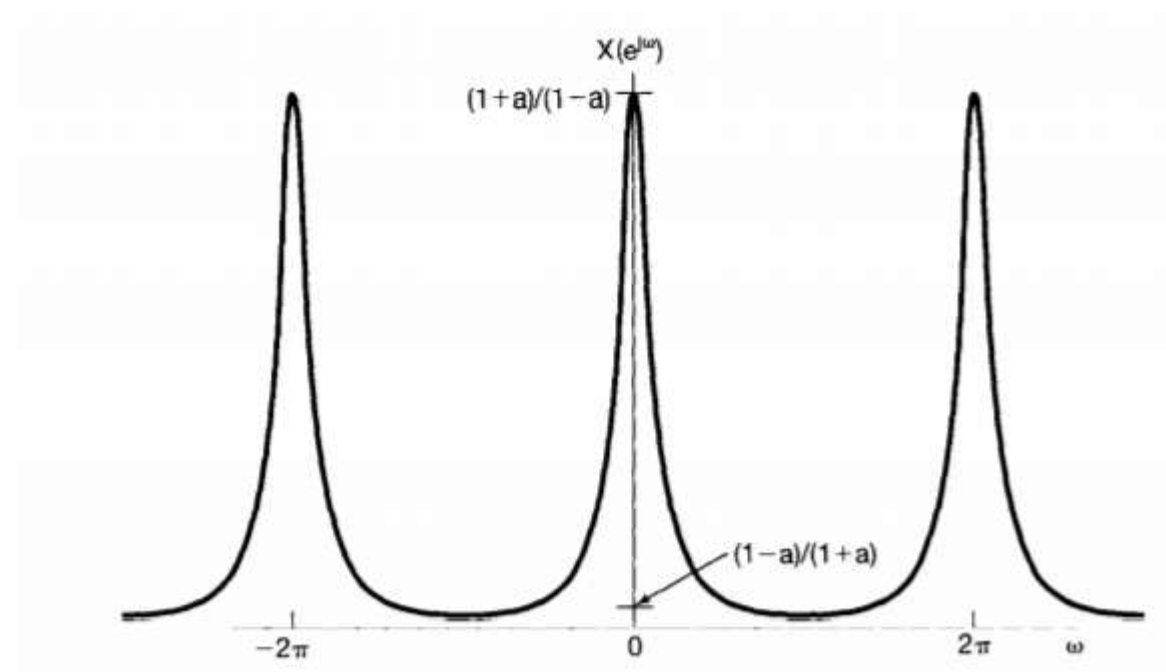
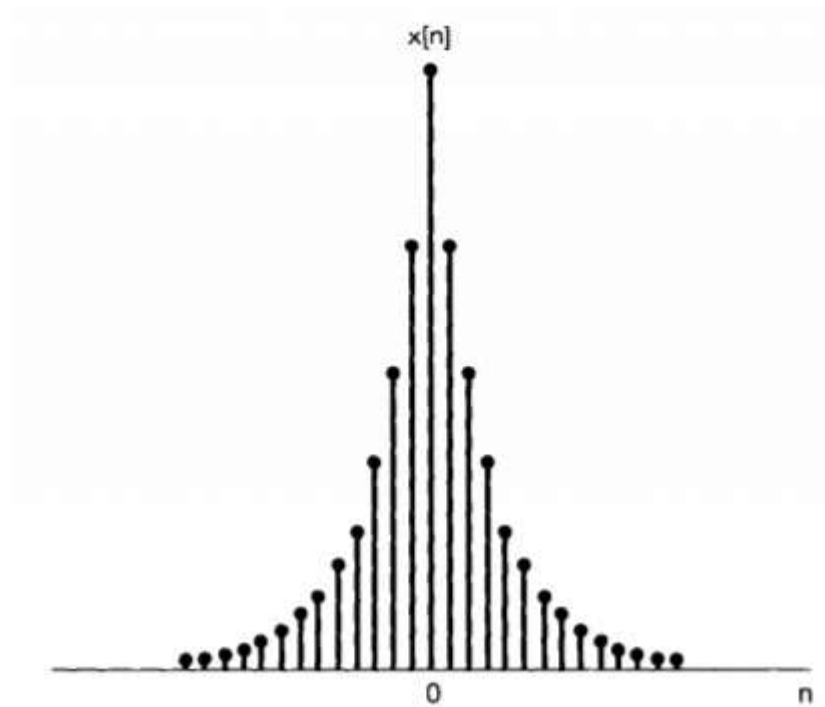
➤ We know, $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$
where, $x[n] = a^{|n|}$, $|a| < 1$

$$x[n] = \begin{cases} a^{-n}, & n < 0 \\ a^n, & n \geq 0 \end{cases}$$

➤ Therefore, $X(e^{j\omega}) = \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n} + \sum_{n=0}^{\infty} a^n e^{-j\omega n}$

➤ Let $m = -n$ in first summation, then

$$\begin{aligned} X(e^{j\omega}) &= \sum_{m=1}^{\infty} (ae^{j\omega})^m + \sum_{n=0}^{\infty} (ae^{-j\omega})^n \\ &= \frac{ae^{j\omega}}{1-ae^{j\omega}} + \frac{1}{1-ae^{-j\omega}} \\ &= \frac{1-a^2}{1-2a\cos\omega+a^2} \end{aligned}$$

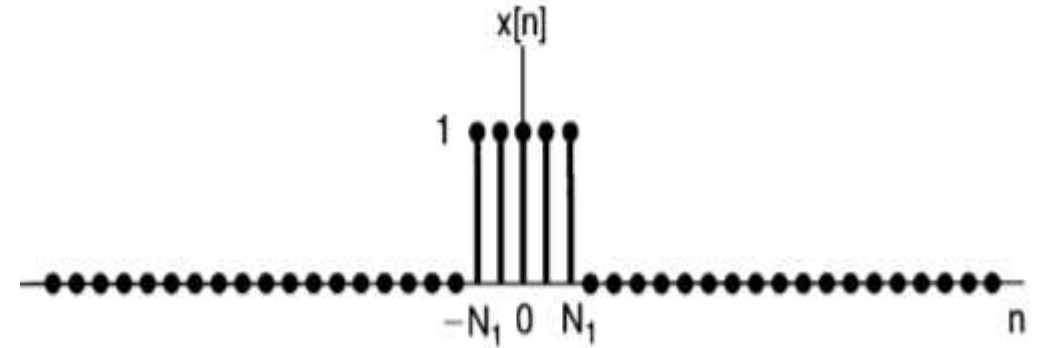


3) Solution:

➤ Given,

$$x[n] = \begin{cases} 1, & |n| \leq N_1 \\ 0, & |n| > N_1 \end{cases}$$

➤ Fig. shows the graph of the signal $x[n]$.



➤ The DTFT of the signal is given as,

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \\ &= \sum_{n=-N_1}^{N_1} 1 e^{-j\omega n} \end{aligned}$$

➤ Let, $m = n + N_1$ then we have

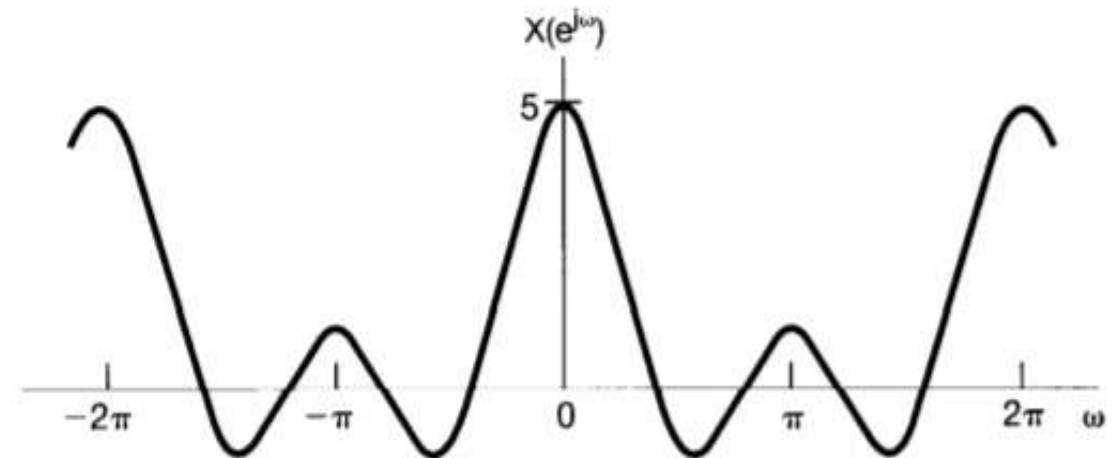
$$\begin{aligned} X(e^{j\omega}) &= \sum_{m=0}^{2N_1} e^{-j\omega(m-N_1)} = e^{j\omega N_1} \sum_{m=0}^{2N_1} (e^{-j\omega})^m \\ &= e^{j\omega N_1} \left\{ \frac{1 - e^{-j\omega(2N_1+1)}}{1 - e^{-j\omega}} \right\} = \frac{e^{j\omega N_1} - e^{-j\omega 2N_1} e^{-j\omega} e^{j\omega N_1}}{1 - e^{-j\omega}} \end{aligned}$$

➤ Or,

$$\begin{aligned}
 X(e^{j\omega}) &= \frac{e^{j\omega N_1} - e^{-j\omega N_1} e^{-j\omega}}{1 - e^{-j\omega}} \\
 &= \frac{e^{-j\frac{\omega}{2}} \{e^{j\omega N_1} e^{j\frac{\omega}{2}} - e^{-j\omega N_1} e^{-j\frac{\omega}{2}}\}}{e^{-j\frac{\omega}{2}} \{e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}\}} \\
 &= \frac{\{e^{j\omega(N_1 + \frac{1}{2})} - e^{-j\omega(N_1 + \frac{1}{2})}\} / 2j}{\{e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}\} / 2j}
 \end{aligned}$$

➤ Therefore, $X(e^{j\omega}) = \frac{\sin\{\omega(N_1 + \frac{1}{2})\}}{\sin(\frac{\omega}{2})}$

➤ Fig. shows the spectrum for $N_1 = 2$.



2. Properties of DTFT:

- The DTFT properties are useful in reducing the complexity in the evaluation of transforms and inverse transforms. Let $x[n]$ and $X(e^{j\omega})$ are DTFT pair and can be written as

$$\begin{aligned} FT\{x[n]\} &= X(e^{j\omega}) \\ FT^{-1}\{X(e^{j\omega})\} &= x[n] \\ x[n] &\xleftrightarrow{FT} X(e^{j\omega}) \end{aligned}$$

a. Linearity:

- If

$$\begin{aligned} x_1[n] &\xleftrightarrow{FT} X_1(e^{j\omega}) \\ x_2[n] &\xleftrightarrow{FT} X_2(e^{j\omega}) \end{aligned}$$

$$\text{then } \{a x_1[n] + b x_2[n]\} \xleftrightarrow{FT} a X_1(e^{j\omega}) + b X_2(e^{j\omega})$$

Proof :

➤ We have, $FT\{a x_1[n] + b x_2[n]\} = \sum_{n=-\infty}^{\infty} \{a x_1[n] + b x_2[n]\} e^{-j\omega n}$

$$= a \left\{ \sum_{n=-\infty}^{\infty} x_1[n] e^{-j\omega n} \right\} + b \left\{ \sum_{n=-\infty}^{\infty} x_2[n] e^{-j\omega n} \right\}$$
$$= a X_1(e^{j\omega}) + b X_2(e^{j\omega})$$

➤ Therefore, $FT\{a x_1[n] + b x_2[n]\} \xleftrightarrow{FT} a X_1(e^{j\omega}) + b X_2(e^{j\omega})$

b. Periodicity:

➤ The DTFT is always periodic in ω with period 2π . That is,

$$X(e^{j\omega+2\pi}) = X(e^{j\omega})$$

This is in contrast (or opposite) to the CTFT, which is in general not periodic.

Proof :

➤ We have,

$$\begin{aligned} X(e^{j\omega+2\pi}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega+2\pi)n} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} e^{-j2\pi n} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \\ &= X(e^{j\omega}) \end{aligned}$$

c. Time shifting:

➤ If

$$x[n] \xleftrightarrow{FT} X(e^{j\omega})$$

then

$$x[n - n_0] \xleftrightarrow{FT} e^{-j\omega n_0} X(e^{j\omega})$$

Proof:

➤ We know, $FT\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

Then $FT\{x[n - n_0]\} = \sum_{n=-\infty}^{\infty} x[n - n_0] e^{-j\omega n}$

➤ Let, $n - n_0 = m$ in RHS of above eq., then

$$FT\{x[n - n_0]\} = \sum_{m=-\infty}^{\infty} x[m] e^{-j\omega(m+n_0)}$$

$$= \left\{ \sum_{m=-\infty}^{\infty} x[m] e^{-j\omega m} \right\} e^{-j\omega n_0}$$

$$= \underset{FT}{X(e^{j\omega})} e^{-j\omega n_0}$$

➤ Therefore, $x[n - n_0] \xleftrightarrow{FT} e^{-j\omega n_0} X(e^{j\omega})$

- This result shows that delaying a signal by n_0 units does not change its amplitude spectrum but the phase spectrum is changed by $-\omega n_0$. This added phase is a linear function of ω with slope n_0 . Therefore time delay in a signal causes linear phase shift in its spectrum.

d. Frequency Shifting:

➤ If $x[n] \xleftrightarrow{FT} X(e^{j\omega})$
then $x[n]e^{j\omega_0 n} \xleftrightarrow{FT} X(e^{j(\omega-\omega_0)})$

Proof:

- We have,
- $$\begin{aligned} FT \{x[n]\} &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \\ FT \{x[n]e^{j\omega_0 n}\} &= \sum_{n=-\infty}^{\infty} \{x[n]e^{j\omega_0 n}\} e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega-\omega_0)n} \\ &= X(e^{j(\omega-\omega_0)}) \end{aligned}$$
- Therefore, $x[n]e^{j\omega_0 n} \xleftrightarrow{FT} X(e^{j(\omega-\omega_0)})$
- The multiplication of sequence $x[n]$ is equivalent to a frequency translation of the spectrum $X(e^{j\omega})$ by ω_0 (i.e. $\omega = \omega_0$). Since the spectrum $X(e^{j\omega})$ is 2π periodic, the shift ω_0 applies to the spectrum of the signal in every period.

e. Time Reversal:

- If a DT signal $x[n]$ is folded in time about the origin then its amplitude or magnitude spectrum remains unchanged but a phase spectrum undergoes a change in sign.

➤ That is , if

$$x[n] \xleftrightarrow{FT} X(e^{j\omega})$$

then

$$x[-n] \xleftrightarrow{FT} X(e^{-j\omega})$$

Proof:

➤ We have,

$$FT \{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$
$$FT \{x[-n]\} = \sum_{n=-\infty}^{\infty} x[-n] e^{-j\omega n} \quad \text{.....i}$$

- Put $m = -n$ in eq.i, we get

$$\begin{aligned} FT \{x[-n]\} &= \sum_{m=-\infty}^{\infty} x[m] e^{j\omega m} \\ &= \sum_{m=-\infty}^{\infty} x[m] e^{-j(-\omega)m} \\ &= X(e^{-j\omega}) \end{aligned}$$

➤ Therefore,

$$x[-n] \xleftrightarrow{FT} X(e^{-j\omega})$$

f. Conjugation and Conjugate Symmetry :

➤ If $x[n] \xleftrightarrow{FT} X(e^{j\omega})$
then $x^*[n] \xleftrightarrow{FT} X^*(e^{-j\omega})$

Proof:

➤ We have, $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

➤ Taking complex conjugate on both sides ,we get

$$[X(e^{j\omega})]^* = [\sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}]^*$$
$$X^*(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x^*[n] e^{j\omega n}$$

➤ Let $\omega = -\omega$, then

$$X^*(e^{-j\omega}) = \sum_{n=-\infty}^{\infty} x^*[n] e^{-j\omega n}$$
$$X^*(e^{-j\omega}) = FT\{x^*[n]\}$$

➤ Therefore, $x^*[n] \xleftrightarrow{FT} X^*(e^{-j\omega})$

g. Convolution:

➤ If $x[n] \xleftrightarrow{FT} X(e^{j\omega})$
and $h[n] \xleftrightarrow{FT} H(e^{j\omega})$
then $y[n] = x[n] * h[n] \xleftrightarrow{FT} Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$
where $x[n]$ = input signal to a LTI system
 $h[n]$ = impulse response of the LTI system , and
 $y[n]$ = output of the LTI system

Proof:

➤ We know, the convolution formula is

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

This convolution equation is called **convolution sum**.

➤ We have the DTFT of $y[n]$ is

$$\begin{aligned} FT \{y[n]\} &= \sum_{n=-\infty}^{\infty} y[n] e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \{x[n] * h[n]\} e^{-j\omega n} \end{aligned}$$

➤ Or

$$\begin{aligned}
 FT \{y[n]\} &= \sum_{n=-\infty}^{\infty} \left\{ \sum_{k=-\infty}^{\infty} x[k] h[n-k] \right\} e^{-j\omega n} \\
 &= \sum_{k=-\infty}^{\infty} x[k] \left\{ \sum_{n=-\infty}^{\infty} h[n-k] e^{-j\omega n} \right\}
 \end{aligned}$$

➤ From time shifting property, we know

$$\begin{aligned}
 FT \{h[n-k]\} &= \sum_{n=-\infty}^{\infty} h[n-k] e^{-j\omega n} \\
 &= e^{-j\omega k} H(e^{j\omega})
 \end{aligned}$$

➤ Therefore,

$$FT \{y[n]\} = H(e^{j\omega}) \left\{ \sum_{k=-\infty}^{\infty} x[k] e^{-j\omega k} \right\}$$

$$FT \{y[n]\} = H(e^{j\omega}) \underset{FT}{X(e^{j\omega})} = Y(e^{j\omega})$$

➤ Thus

$$y[n] = x[n] * h[n] \longleftrightarrow Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$$

➤ This shows that the convolution of two signals in time domain is equal the multiplication in their Fourier transforms, i.e., multiplication in frequency domain.

h. Multiplication of Two Sequences (Windowing Theorem):

➤ If

$$\begin{array}{ccc} x_1[n] & \xleftrightarrow{FT} & X_1(e^{j\omega}) \\ x_2[n] & \xleftrightarrow{FT} & X_2(e^{j\omega}) \end{array}$$

then

$$x_3[n] = x_1[n] x_2[n] \xleftrightarrow{FT} X_3(e^{j\omega}) = \frac{1}{2\pi} \{X_1(e^{j\omega}) * X_2(e^{j\omega})\}$$

Proof:

➤ We know,

$$\begin{aligned} FT \{x_3[n]\} &= \sum_{n=-\infty}^{\infty} x_3[n] e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \{x_1[n] x_2[n]\} e^{-j\omega n} \end{aligned}$$

where,

$$x_1[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\lambda}) e^{j\lambda n} d\lambda, \text{ then}$$

$$\begin{aligned} FT \{x_3[n]\} &= \sum_{n=-\infty}^{\infty} \left\{ \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\lambda}) e^{j\lambda n} d\lambda \right\} x_2[n] e^{-j\omega n} \\ &= \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\lambda}) \sum_{n=-\infty}^{\infty} x_2[n] e^{-j\omega n} e^{j\lambda n} d\lambda \end{aligned}$$

➤ Or

$$\begin{aligned}
 FT \{x_3[n]\} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(e^{j\lambda}) \left\{ \sum_{n=-\infty}^{\infty} x_2[n] e^{-j(\omega-\lambda)n} \right\} d\lambda \\
 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(e^{j\lambda}) X_2(e^{j(\omega-\lambda)}) d\lambda \\
 &= \frac{1}{2\pi} \{X_1(e^{j\omega}) * X_2(e^{j\omega})\} = X_3(e^{j\omega})
 \end{aligned}$$

➤ Therefore, $x_3[n] = x_1[n] x_2[n] \xleftrightarrow{FT} X_3(e^{j\omega}) = \frac{1}{2\pi} \{X_1(e^{j\omega}) * X_2(e^{j\omega})\}$

➤ Since there is periodic convolution in frequency domain, this property is also called **frequency convolution property**.

i. Parseval's Theorem:

- It states that the total energy of a discrete-time signal $x[n]$ may be obtained from the knowledge of its DTFT.

➤ That is, if
$$x[n] \xleftrightarrow{FT} X(e^{j\omega})$$
 then
$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

Proof:

➤ We have,
$$\begin{aligned} E = \sum_{n=-\infty}^{\infty} |x[n]|^2 &= \sum_{n=-\infty}^{\infty} \{x[n] x^*[n]\} \\ &= \sum_{n=-\infty}^{\infty} x[n] \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) e^{-j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) \left\{ \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right\} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) X(e^{j\omega}) d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega \end{aligned}$$

➤ Therefore,
$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

- Since, $|X(e^{j\omega})|^2$ represents the distribution of energy as a function of ω (or frequency), it is known as **energy density spectrum** of DT signal $x[n]$.

j. Differentiation in Frequency: Multiplication by n

➤ If

$$x[n] \xleftrightarrow{FT} X(e^{j\omega})$$

then

$$n x[n] \xleftrightarrow{FT} j \frac{d\{X(e^{j\omega})\}}{d\omega}$$

Proof:

- We know, $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$
- Differentiating on both sides, we get

$$\frac{d\{X(e^{j\omega})\}}{d\omega} = \sum_{n=-\infty}^{\infty} -jn x[n] e^{-j\omega n}$$

$$j \frac{d\{X(e^{j\omega})\}}{d\omega} = -(j)^2 \left\{ \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right\} n$$

➤ Or

$$j \frac{d\{X(e^{j\omega})\}}{d\omega} = n \text{ FT}\{x[n]\}$$

➤ Therefore,

$$n x[n] \xleftrightarrow{FT} j \frac{d\{X(e^{j\omega})\}}{d\omega}$$

❑ Discrete-Time System Properties:

1. System Definition:

- A discrete-time system is defined as a device or algorithm that operates on discrete-time signal(s) called input or excitation, according to a some well defined rule(s), to produce another discrete-time signal(s) called output or response.

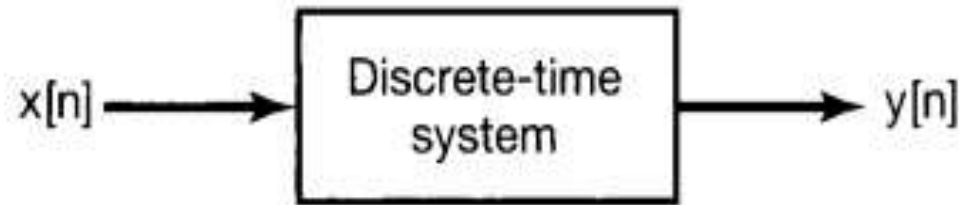


Fig: A discrete-time system

- We say that the input signal $x[n]$ is transformed by the system to the output signal $y[n]$ and express the general relationship between $x[n]$ and $y[n]$ as

$$y[n] = T\{x[n]\}$$

.....1

where,

$x[n]$ = input signal

$y[n]$ = output signal,

and

T = transformation or operator (i. e. , system)

➤ The input-output relationship of equation 1 is written as

$$x[n] \xrightarrow{T} y[n]$$

2. Properties of System:

➤ There are number of system properties of both CT and DT systems, and **the systems are classified** on the basis of **system properties** that they satisfy.

1. Systems with and without memory (memory)
2. Invertibility and inverse systems (invertibility)
3. Causal and noncausal systems (causality)
4. Time variant and time invariant systems (time-invariance)
5. Stable and unstable systems (stability)
6. Linear and nonlinear systems (linearity)

a. Systems with and without memory (memory):

➤ A system is said to be **memoryless** if its output signal value depends on the present value of input signal but does not depend on past and future values. A memoryless system is also called **static** system.

➤ For example,

i. A **resistor** system

$$y(t) = R x(t)$$

where, $x(t)$ = current

$y(t)$ = voltage

and R = resistor

ii. An identity system

$$y(t) = x(t) \quad , \text{ for CT system}$$

$$y[n] = x[n] \quad , \text{ for DT system}$$

iii. $y(t) = 2 x(t) - x^2(t)$

➤ A system has **memory** if its output signal value depends upon the past (also may be future) values of the input signal. A system having memory is also known as **dynamic** system.

➤ For example

i. A **capacitor** system

$$y(t) = \frac{1}{C} \int_{-\infty}^t x(\tau) d\tau \quad \text{CT system}$$

ii. An **accumulator or summer** system

$$y[n] = \sum_{k=-\infty}^n x[k] \quad \text{DT system}$$

2) Invertibility and inverse system:

➤ A system is said to be **invertible** if distinct inputs leads to distinct outputs. In other words, a system is said to be invertible system if the input of the system is or can be recovered from the output signal.

➤ If a system is invertible then there exists an inverse system that when cascaded with the original system which yields an output signal equal to the input signal.

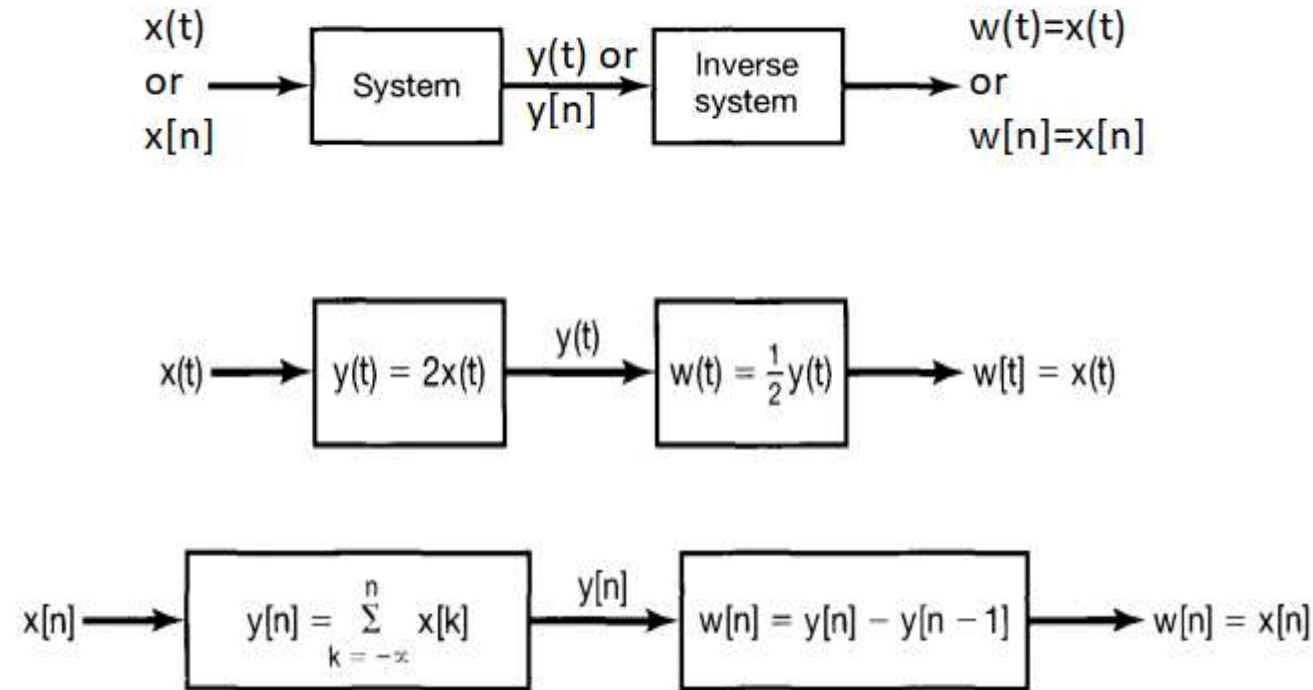


Fig.: Concept of an inverse system

- The system which produces zero output signal $\{y[n] = 0 \text{ or } y(t) = 0\}$ for any input signal and if we cannot determine the sign of the input from the knowledge of the output $\{y(t) = x^2(t) \text{ or } y[n] = x^2[n]\}$, are **non-invertible systems**.

3) Causal and noncausal system:

- A **causal system** is one whose output at any time depends only on the present and past inputs but does not depend on the future inputs. All the real time systems are causal systems because the systems can not know the future values of the input signal.
- The causal systems are physically realizable.
- For example,
 - i. $y(t) = \frac{1}{C} \int_{-\infty}^t x(\tau) d\tau$; output depends on present and past values of inputs only
 - ii. $y[n] = x[n] - x[n - 1]$
 - iii. $y[n] = 2 a x[n]$
 - iv. $v(t) = R i(t)$
- A system is **noncausal** if its output at any time depends on the future values of the input signal.
- For example,
 - i. $y(t) = 2x(t) + x(t - 2) + x(t + 1)$
 - ii. $y[n] = x[n] + 2x[n - 1] + 4x[n + 2]$

4) Time-variant and time invariant system:

- A system is said to be **time invariant** or **shift invariant** if its input-output characteristic do not vary with time. That is,

$$y(t) = T\{x(t)\} \quad \text{for CT system}$$
$$y[n] = T\{x[n]\}$$

- Thus, for a time invariant system, a time shift in the input signal results in the corresponding time shift in the output signal.

$$y(t - t_0) = T\{x(t - t_0)\}$$
$$y[n - k] = T\{x[n - k]\}$$

- If a system does not satisfy the above equations, the system is said to be **time variant** or **shift variant** system.

■ Examples:

- 1) Determine whether the system given below are time variant or time invariant.

a) $y(t) = \sin\{x(t)\}$	b) $y[n] = nx[n]$
c) $y[n] = x[n] \cos \omega_0 n$	d) $y[n] = x[-n]$
e) $y[n] = x[n] + x[n - 1]$	

Solution:

a) Given, $y(t) = \sin\{x(t)\}$

➤ First, we delay the input in t_0 seconds in time , we get

$$y(t, t_0) = \sin\{x(t - t_0)\} \quad \text{.....i}$$

➤ Again, we delay output by t_0 seconds in time ,we get

$$y(t - t_0) = \sin\{x(t - t_0)\} \quad \text{.....ii}$$

➤ Since, $y(t, t_0) = y(t - t_0)$, the system is time invariant system.

b) Given, $y[n] = nx[n]$

➤ First, we delay the input in k units in time , we get

$$y[n, k] = n x[n - k] \quad \text{.....i}$$

➤ Again, we delay output by k units in time ,we get

$$y[n - k] = (n - k) x[n - k] \quad \text{.....ii}$$

➤ Since, $y[n, k] \neq y[n - k]$, the system is time variant system.

5) Stable and unstable system:

- **Bounded and unbounded signal:**

- If a signal $x(t)$ is bounded(finite), its magnitude is always a finite value and we can write
$$|x(t)| \leq M$$

where, $M =$ positive real finite number (maximum value)

- For example, $|A \cos \omega_0 t| \leq A$. That is, a cosine wave is a bounded signal since its magnitude is always less than or equal to its peak or maximum magnitude.

- **Definition:**

- A system is said to be **bounded input-bounded output(BIBO) stable** if and only if every bounded(or finite) input produces a bounded output. The output of bounded system does not diverge or does not grow unreasonably large.

- If an input signal and output signal are bounded then there exists some finite numbers, say M_x and M_y , such that

$$|x(t)| \leq M_x < \infty \quad \text{and} \quad |y(t)| \leq M_y < \infty \quad \text{for CT system}$$

$$|x[n]| \leq M_x < \infty \quad \text{and} \quad |y[n]| \leq M_y < \infty \quad \text{for DT system}$$

- If for some bounded input signal, if the output is unbounded(infinite), the system is said to be **unstable system**.

6) Linear and nonlinear system:

➤ A **linear system** is one that satisfies the superposition principle. That is, a linear system may be defined as one whose response to the sum of weighted inputs is equal to the sum of weighted responses.

➤ A system is linear if and only if

$$\begin{aligned} T\{a x_1(t) + b x_2(t)\} &= a T\{x_1(t)\} + b T\{x_2(t)\} \\ &= a y_1(t) + b y_2(t) \end{aligned} \quad \text{for CT systemi}$$

$$\begin{aligned} T\{a x_1[n] + b x_2[n]\} &= a T\{x_1[n]\} + b T\{x_2[n]\} \\ &= a y_1[n] + b y_2[n] \end{aligned} \quad \text{for DT systemii}$$

➤ A **nonlinear system** does not satisfy superposition principle and hence above equations.

➤ **Examples:**

1) Determine whether the systems described by following input-output equations are linear or nonlinear.

a) $y(t) = t x(t)$

b) $y[n] = x[n^2]$

c) $y(t) = x^2(t)$

d) $y[n] = A x[n] + B$

e) $y[n] = e^{x[n]}$

Solution:

a) Given, $y(t) = t x(t)$

➤ The output of the system for two input signals $x_1(t)$ and $x_2(t)$ are:

$$y_1(t) = T\{x_1(t)\} = t x_1(t), \text{ and}$$

$$y_2(t) = T\{x_2(t)\} = t x_2(t)$$

Thus, the sum of weighted outputs is

$$y'(t) = a t x_1(t) + b t x_2(t) \quad \text{.....i}$$

➤ Let $x_3(t)$ be the linear combination of two inputs $x_1(t)$ and $x_2(t)$. That is,

$$x_3(t) = a x_1(t) + b x_2(t)$$

If $x_3(t)$ be the input to the system, then the response of the sum of weighted inputs is

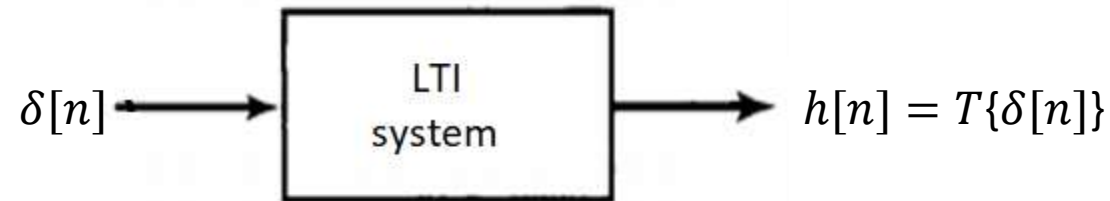
$$y''(t) = T\{x_3(t)\} = t x_3(t) = a t x_1(t) + b t x_2(t) \quad \text{.....ii}$$

➤ Since, $y'(t) = y''(t)$, the system is linear system.

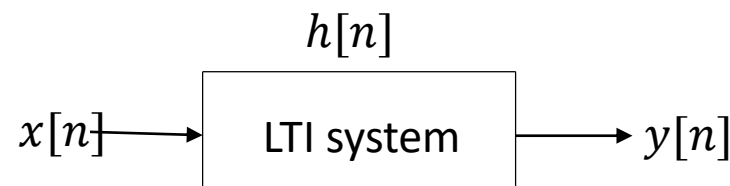
❑ Linear-Time Invariant (LTI) System, Convolution Sum, Properties of LTI Systems:

1. Introduction:

- A time invariant system which holds the linearity property in its input-output relationship is called **linear time invariant(LTI) system**. A LTI system can be completely characterized by its **impulse response**.
- **Impulse response** of the LTI system is the response of the system when it is excited by impulse function.

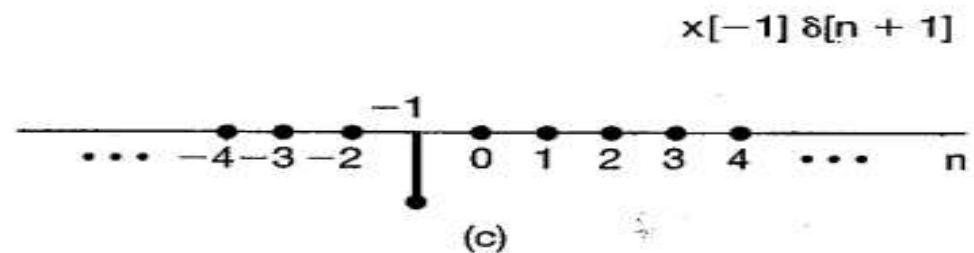
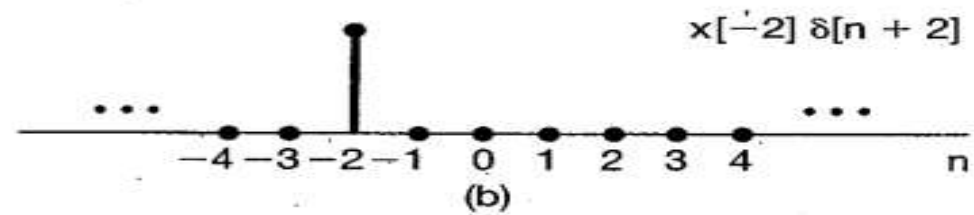
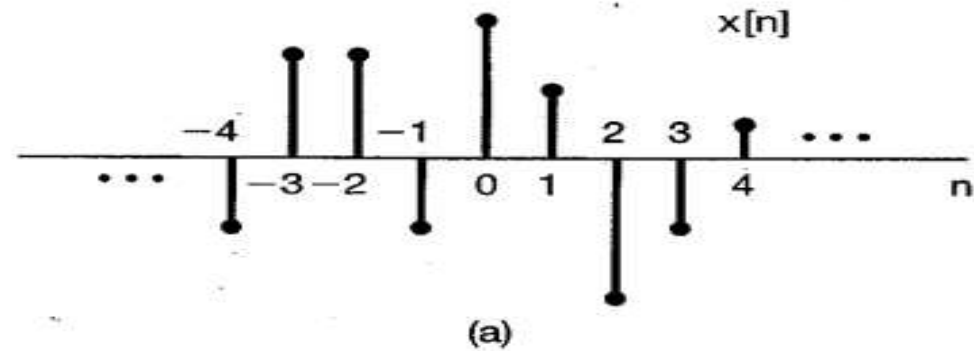


- In the above block diagram, $h[n]$ is the DT impulse response.
- Thus, a LTI system can be represented as shown below:



2. Representation or Resolution of Discrete-Time Signals in terms of Impulses:

- A discrete-time (DT) signal $x[n]$ can be resolved into the sum of weighted impulse sequences. The concept is illustrated below:



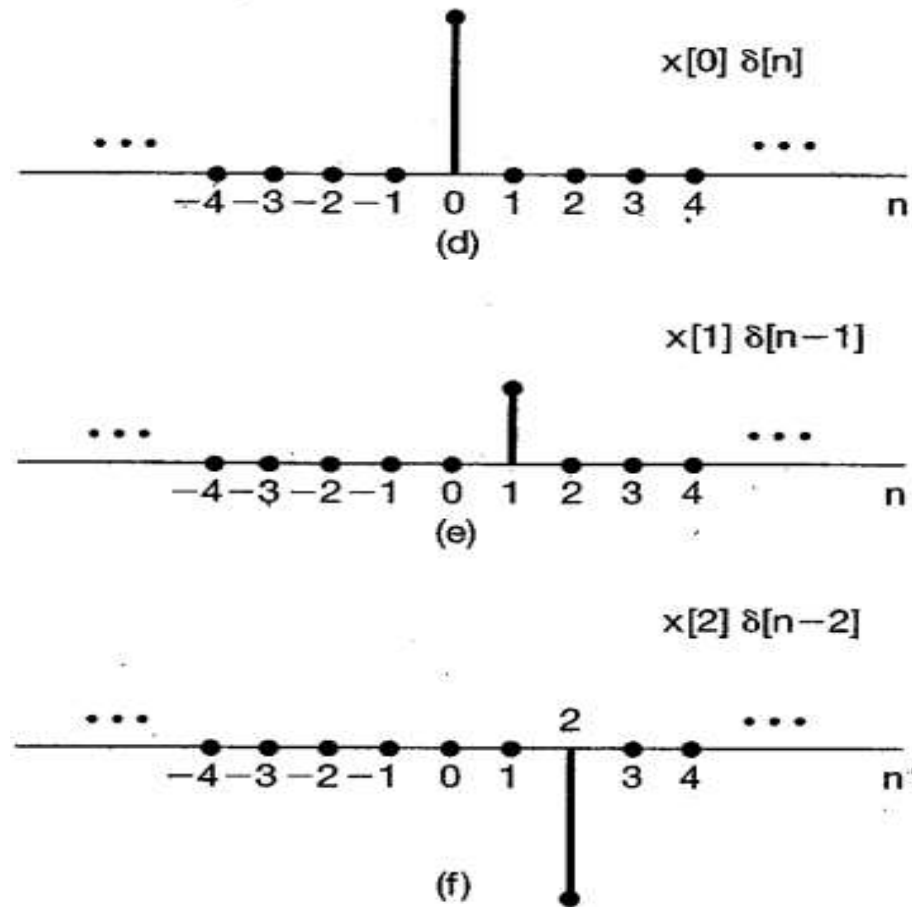


Fig: Decomposition of a DT signal into a weighted sum of shifted impulses

➤ From above fig. , we can write

$$x[-2]\delta[n+2] = \begin{cases} x[-2], & n = -2 \\ 0 & , n \neq -2 \end{cases}$$

$$x[-1]\delta[n+1] = \begin{cases} x[-1], & n = -1 \\ 0 & , n \neq -1 \end{cases}$$

$$x[0]\delta[n] = \begin{cases} x[0], & n = 0 \\ 0 & , n \neq 0 \end{cases}$$

$$x[1]\delta[n-1] = \begin{cases} x[1], & n = 1 \\ 0 & , n \neq 1 \end{cases}$$

➤ In general, we can write

$$x[n] = \cdots + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \cdots$$

➤ Therefore, $x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$ 1

Equation (1) corresponds to a representation of an arbitrary sequence $x[n]$ as a linear combination of shifted impulses $\delta[n-k]$ where the weights in this linear combination are $x[k]$.

3. Response of LTI Systems to Arbitrary Inputs: The Convolution Sum

- Any arbitrary input signal $x[n]$ can be expressed in terms of weighted impulses as

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k] \quad \text{.....1}$$

- Now, if the LTI system is excited by $x[n]$, then the output $y[n]$ can be expressed as

$$\begin{aligned} y[n] &= T\{x[n]\} \\ y[n] &= T\{\sum_{k=-\infty}^{\infty} x[k] \delta[n - k]\} \end{aligned} \quad \text{.....2}$$

- Since, the system is linear, we can write

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] T\{\delta[n - k]\} \quad \text{.....3}$$

- Again, since the system is time-invariant, we can write

$$\begin{aligned} h[n] &= T\{\delta[n]\} \\ \text{then } h[n - k] &= T\{\delta[n - k]\} \end{aligned} \quad \text{.....4}$$

- From equations (3) and (4), we can write

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] \quad \text{.....5}$$

Equation (5) gives the response $y[n]$ of the LTI system as a function of input signal $x[n]$ and the unit impulse response $h[n]$, called **convolution sum**.

- Equation (5) can be written as $y[n] = x[n] * h[n]$ 6

- Suppose we wish to compute output of the system at time $n = n_0$, then we have

$$y[n_0] = \sum_{k=-\infty}^{\infty} x[k]h[n_0 - k] \quad \text{.....7}$$

This computation can be done in four steps:

1. **Folding**: Fold $h[k]$ about $k=0$ to obtain $h[-k]$.
 2. **Shifting**: shift $h[-k]$ by n_0 to the right (left) if n_0 is positive (negative) to obtain $h[n_0 - k]$.
 3. **Multiplication**: Multiply $x[k]$ by $h[n_0 - k]$ to obtain the product $v_{n_0}[k] = x[k]h[n_0 - k]$.
 4. **Summation**: Sum all the values of the product sequence $v_{n_0}[k]$ to obtain the output at time $n = n_0$.
- Note that steps 2 through 4 in the summary must be repeated, for all the possible time shift $-\infty < n < \infty$.

Example: The impulse response of a LTI system is

$$h[n] = \{1, 2, 1, -1\}$$

Determine the response of the system to the input

$$x[n] = \{\mathbf{1}, 2, 3, 1\}$$

where bold number represents value at origin ($n = 0$)

Solution:

1. Graphical Method:

➤ Given, $h[n] = \{1, \mathbf{2}, 1, -1\}$, impulse response, and

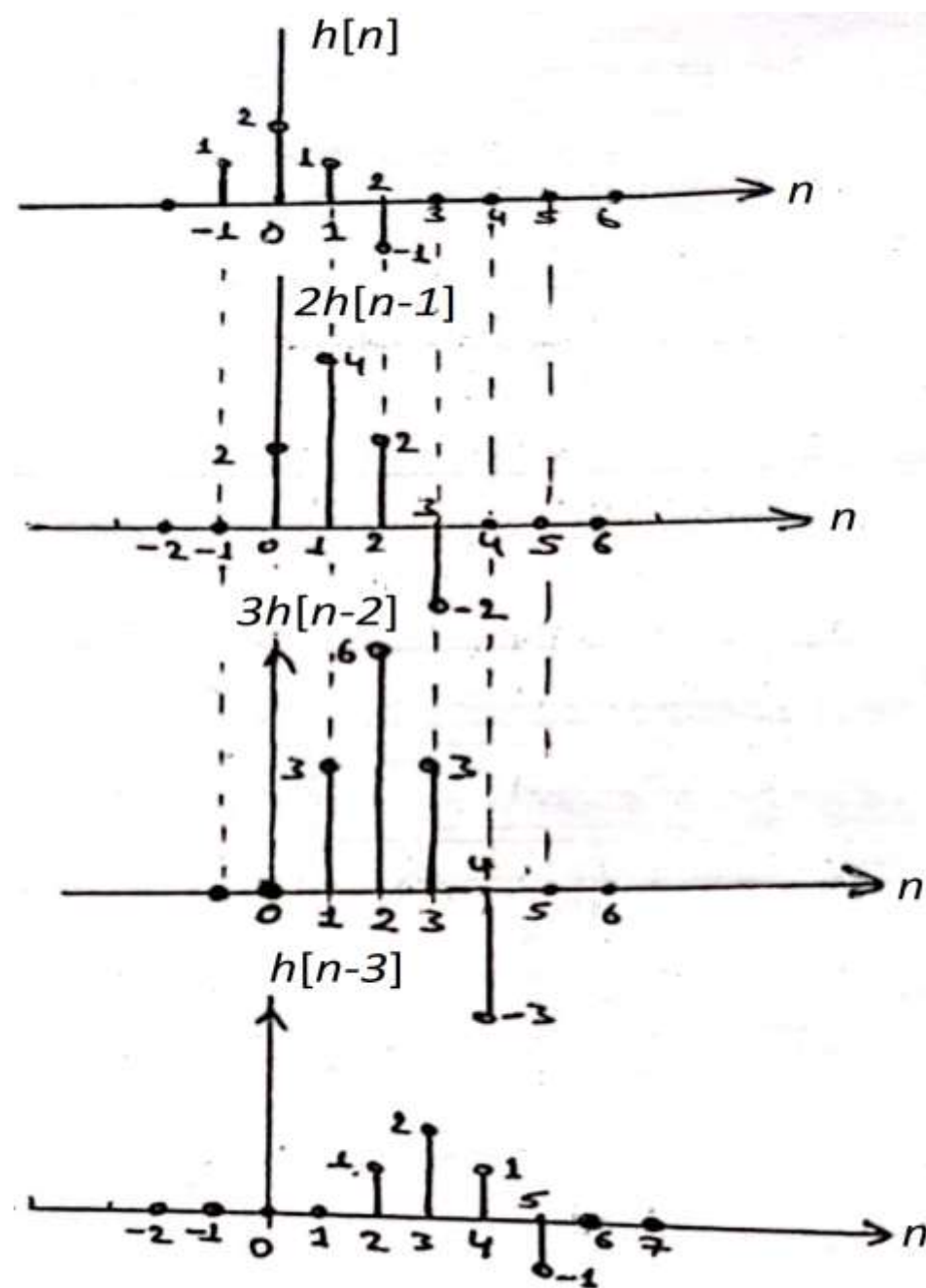
$x[n] = \{\mathbf{1}, 2, 3, 1\}$, input signal

➤ The output $y[n]$ is given by the convolution sum as

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k]h[n-k] \\ &= \cdots + x[-1]h[n+1] + x[0]h[n] + x[1]h[n-1] + x[2]h[n-2] + \\ &\quad x[3]h[n-3] + x[4]h[n-4] + \cdots \\ &= 0 + x[0]h[n] + x[1]h[n-1] + x[2]h[n-2] + x[3]h[n-3] + 0 \end{aligned}$$

➤ Therefore, $y[n] = h[n] + 2h[n-1] + 3h[n-2] + h[n-3] \dots i$

➤ Now, we draw the individual signal of $h[n]$, $2h[n-1]$, $3h[n-2]$, $h[n-3]$, and add them to obtain output $y[n]$ as illustrated in the following figures:



➤ Therefore, the output is

$$y[n] = \{1, 4, 8, 8, 3, -2, -1\}$$

2. Mathematical Method:

➤ The output $y[n]$ is given by the convolution sum as

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

➤ Now, $y[0] = \sum_{k=-\infty}^{\infty} x[k]h[-k]$
 $= \cdots + x[-1]h[1] + x[0]h[0] + x[1]h[-1] + x[2]h[-2] + \cdots$
 $= 0 + 1 \times 2 + 2 \times 1 + 0 = 4$

$$\begin{aligned} y[1] &= \sum_{k=-\infty}^{\infty} x[k]h[1-k] \\ &= \cdots + x[0]h[1] + x[1]h[0] + x[2]h[-1] + \cdots \\ &= 1 \times 1 + 2 \times 2 + 3 \times 1 = 8 \end{aligned}$$

$$\begin{aligned} y[2] &= \sum_{k=-\infty}^{\infty} x[k]h[2-k] \\ &= \cdots + x[0]h[2] + x[1]h[1] + x[2]h[0] + x[3]h[-1] + \cdots \\ &= 1 \times -1 + 2 \times 2 + 3 \times 2 + 1 \times 1 = 8 \end{aligned}$$

➤ Similarly ,

$$\begin{aligned}
 y[3] &= \sum_{k=-\infty}^{\infty} x[k]h[3-k] \\
 &= \cdots + x[1]h[2] + x[2]h[1] + x[3]h[0] + \cdots \\
 &= 2 \times (-1) + 3 \times 1 + 1 \times 2 = 3 \\
 y[4] &= \sum_{k=-\infty}^{\infty} x[k]h[4-k] \\
 &= \cdots + x[2]h[2] + x[3]h[1] + \cdots \\
 &= 3 \times (-1) + 1 \times 1 = -2 \\
 y[5] &= \sum_{k=-\infty}^{\infty} x[k]h[5-k] \\
 &= \cdots + x[3]h[2] + \cdots \\
 &= 1 \times (-1) = -1 \\
 y[6] &= \sum_{k=-\infty}^{\infty} x[k]h[6-k] \\
 &= \cdots + x[3]h[3] + \cdots \\
 &= 1 \times 0 = 0
 \end{aligned}$$

and $y[n] = 0$ for $n \geq 6$

➤ Again,

$$\begin{aligned}
 y[-1] &= \sum_{k=-\infty}^{\infty} x[k]h[-1-k] \\
 &= \cdots + x[0]h[-1] + \cdots \\
 &= 1 \times 1 = 1
 \end{aligned}$$

➤ Similarly,
$$y[-2] = \sum_{k=-\infty}^{\infty} x[k]h[-2-k]$$

$$= \dots + x[0]h[-2] + \dots$$

$$= 1 \times 0 = 0$$

and $y[n] = 0$ for $n \leq -2$

➤ Therefore, the output is, $y[n] = \{1, 4, 8, 8, 3, -2, -1\}$

3. Overlap and Add Method:

$x[n]$ \ $h[n]$	1	2	1	-1
1	1	2	1	-1
2	2	4	2	-2
3	3	6	3	-3
1	1	2	1	-1

$$y[n] = \{1, 4, 8, 8, 3, -2, -1\}$$

○ **Note:**

• **Determination of length of $y[n]$:**

- Let, length of $x[n] = L_1$ and length of $h[n] = L_2$
then length of $y[n]$, $L = L_1 + L_2 - 1$
- For above example, $L = 4 + 4 - 1 = 7$

• **Range of convoluted signal $y[n]$:**

- It is given by

$$\text{sum of lower limits} \leq n \leq \text{sum of upper limits}$$

$$0 + (-1) \leq n \leq 3 + 2$$

$$-1 \leq n \leq 5$$

- Therefore, output signal ranges from -1 to 5 .

Example: Find the output of the signal $y[n]$ of a relaxed LTI system with impulse response $h[n] = a^n u[n]$, $|a| < 1$, when the input is a unit step sequence.

Solution:

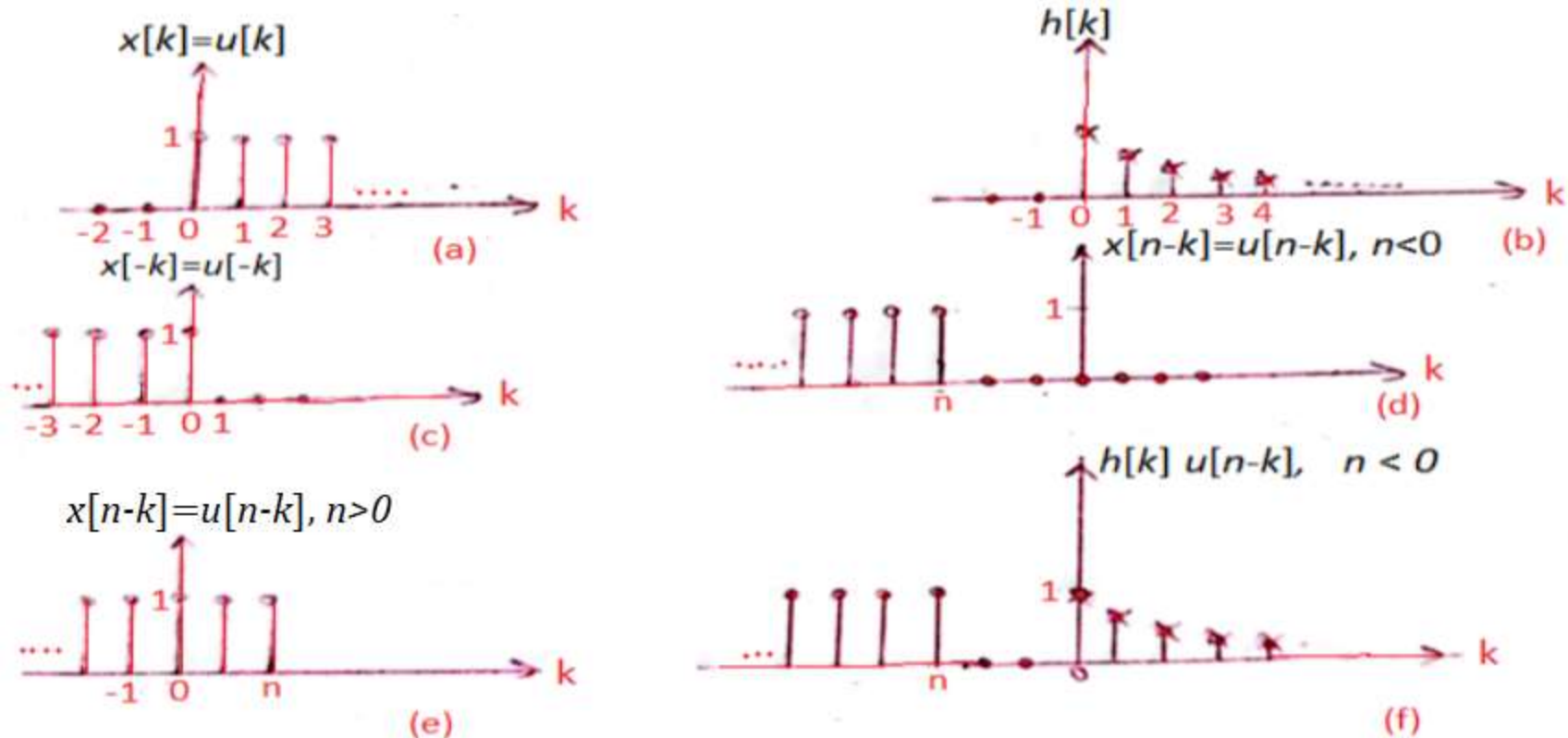
➤ Given, $x[n] = u[n]$ and $u[n] = \begin{cases} 1, n \geq 0 \\ 0, n < 0 \end{cases}$

also, $h[n] = a^n u[n], |a| < 1$

➤ The output LTI system is given by the convolution sum as:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k]$$

Or, $y[n] = \sum_{k=-\infty}^{\infty} h[k] u[n-k]$



- **Interval-1:**

- For $n < 0$, the output $y[n] = 0$. This is because there is no nonzero overlap between the points of $h[k]$ and $x[n - k]$. This is shown in fig(f).

- **Interval-2:**

- Here, $n > 0$, then the output is given by

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] u[n - k]$$

where,
$$h[k] u[n - k] = \begin{cases} a^k, & 0 \leq k \leq n \\ 0, & \text{otherwise} \end{cases}$$

As shown in fig(g), k varies from 0 to n .

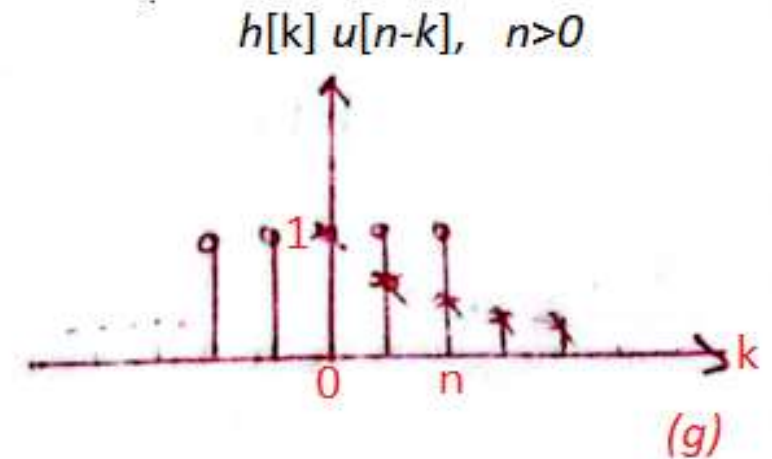
- Therefore, $y[n] = \sum_{k=0}^n a^k = \frac{1-a^{n+1}}{1-a}, n \geq 0$

This is the geometric series and we have,

$$y[n] = 1 + a + a^2 + a^3 + \dots$$

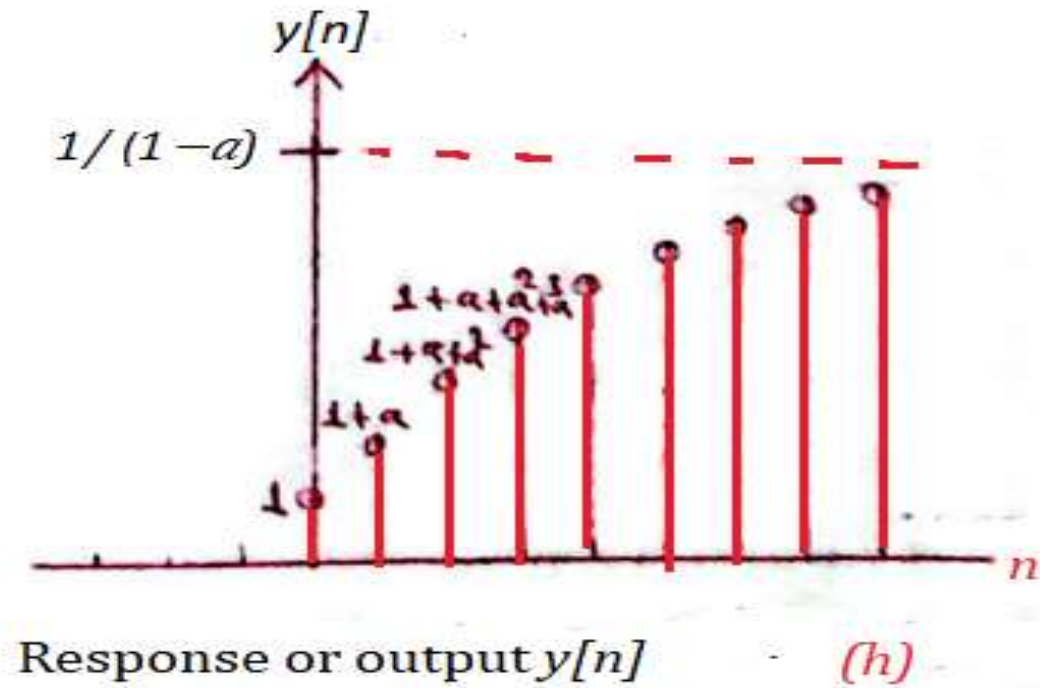
- The final value of $y[n]$ is obtained as $n \rightarrow \infty$.

that is,
$$y[\infty] = \lim_{n \rightarrow \infty} y[n] = \frac{1}{1-a}$$



➤ The output is

$$y[n] = \begin{cases} 0, & n < 0 \\ \frac{1-a^{n+1}}{1-a}, & n \geq 0 \end{cases}$$



4. Properties of LTI System:

➤ We know that the convolution sum is given by

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = x[n] * h[n]$$

Then the properties of LTI systems are given below:

- a. Commutative
 - b. Distributive
 - c. Associative
 - d. Causality
 - e. Stability
 - f. Invertibility
 - g. Unit step response
 - h. Memory
- Note that commutative, distributive and associative properties are also the properties of **convolution**.

a. Commutative:

- A basic property of convolution (either CT or DT) is that it is commutative operation.

That is,
$$y[n] = x[n] * h[n] = h[n] * x[n]$$

Proof:

- We know, $y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$

Let, $n - k = s$, then $k = n - s$

As $k \rightarrow -\infty, s \rightarrow \infty$ and $k \rightarrow \infty, s \rightarrow -\infty$

- Therefore,
$$y[n] = x[n] * h[n] = \sum_{s=-\infty}^{\infty} x[n-s]h[s]$$
$$= \sum_{s=-\infty}^{\infty} h[s]x[n-s]$$

$$y[n] = x[n] * h[n] = h[n] * x[n]$$

b. Distributive:

- Another basic property of convolution is distributive. Specifically, convolution distributive over addition. Then we have,

$$x[n] * \{h_1[n] + h_2[n]\} = x[n] * h_1[n] + x[n] * h_2[n]$$

c. Associative:

- Another important and useful property of convolution is associative. That is,

$$x[n] * \{h_1[n] * h_2[n]\} = \{x[n] * h_1[n]\} * h_2[n]$$

d. Causality:

- A causal system is one whose output depends only on the present and past inputs. In case of LTI systems, ***causality is measured in terms of the impulse response of the LTI system.***

- We know the convolution sum is

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

$$y[n] = \underbrace{\sum_{k=-\infty}^{-1} h[k]x[n-k]}_{(I)} + \underbrace{\sum_{k=0}^{\infty} h[k]x[n-k]}_{(II)}$$

- The (I) terms include future inputs and the (II) includes present and past inputs. If we decompose first term, we get,

$$y[n] = \dots + h[-2]x[n+2] + h[-1]x[n+1]; \text{ thus the inputs are future inputs.}$$

- The LTI system is causal if the first term become zero. For this the impulse response must satisfy the condition

$$h[n] = 0, \text{ for } n < 0.$$

This is both ***necessary and sufficient condition*** for causal LTI system.

e. Stability:

- A system is BIBO stable if every bounded input produces bounded output. ***In LTI system stability is also measured in terms of its impulse response.***
- Consider an input signal $x[n]$ is bounded in magnitude, there exists a constant M_x such that

$$|x[n]| \leq M_x < \infty$$

- Suppose this bounded input is applied to a LTI system, the output is

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

The magnitude of the output is

$$|y[n]| = |\sum_{k=-\infty}^{\infty} h[k]x[n-k]|$$

- But we know,

$$|y[n]| \leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]|$$

Also we know,

$$|x[n-k]| \leq M_x, \text{ therefore we can write}$$

$$|y[n]| \leq M_x \sum_{k=-\infty}^{\infty} |h[k]| \quad \text{.....i}$$

- From eq. i, it is clear that the output is bounded if the ***impulse response of the LTI system is absolutely summable.*** That is, if

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

This is both ***necessary and sufficient condition*** for stability of a LTI system.

f. Invertibility:

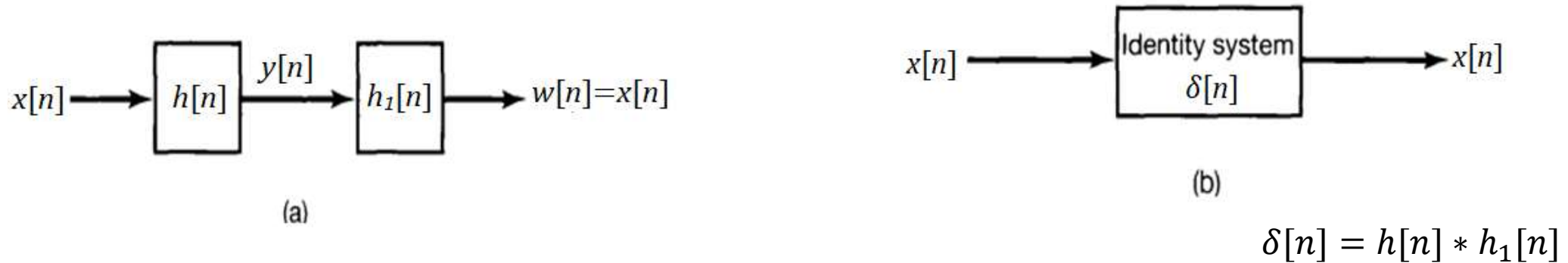


Fig.: An inverse system for continuous-time LTI system

- Consider $h[n]$ be the impulse response of LTI system. Then the LTI system is invertible only when an inverse system exists that when cascaded with the original system, produces an O/P equal to the I/P to the first system.
- In fig., $h_1[n]$ is the inverse system and

$$h[n] * h_1[n] = \delta[n]$$

g. Unit Step Response:

- The unit step response is the response of LTI system when the input is $x[n] = u[n]$.

Therefore, the unit step response of LTI system is given by

$$s[n] = u[n] * h[n] = h[n] * u[n]$$

- That is, $s[n] = \sum_{k=-\infty}^{\infty} h[k]u[n-k] = \sum_{k=-\infty}^n h[k]$ i
- $s[n] = \sum_{k=-\infty}^{n-1} h[k] + h[n]$ ii
- From (i), also we have, $s[n-1] = \sum_{k=-\infty}^{n-1} h[k]$ iii
- From (ii) and (iii), we have $s[n] = s[n-1] + h[n]$
- Therefore, $h[n] = s[n] - s[n-1]$ iv

h. Memory:

- The response LTI system is

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = x[n] * h[n] \quad \text{.....i}$$

- A discrete-time LTI system is **memoryless** if

$$h[n] = 0 \text{ for } n \neq 0 \quad \text{.....ii}$$

and such a memoryless system has the form

$$y[n] = k x[n] \quad \text{.....iii}$$

for some constant k and has the impulse response

$$h[n] = k \delta[n] \quad \text{.....iv}$$

- For $k = 1$ in eq. iv, the system is then called identity system. And we can write the convolution integral formula as

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k] = x[n] * \delta[n]$$

$$\{x[n] * \delta[n] = x[n]\}$$

- If the impulse response $h[n] \neq 0$ for $n \neq 0$, the system is called **dynamic system** or LTI system with **memory**.

❑ Frequency Response of LTI System:

- If $x[n]$ is the input signal and $h[n]$ is the unit impulse response of LTI system then the response or output $y[n]$ of the LTI system is expressed in terms of convolution sum as

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] \quad \text{.....i}$$

$$= x[n] * h[n]$$

- To analyze LTI system, it is convenient to utilize the frequency domain because differential equation and convolution operation in the time domain become algebraic operation in frequency domain.

- Applying convolution property of DTFT in above equation, we get

$$Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$$

and
$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} \quad \dots.i$$

where $H(e^{j\omega})$ is the **frequency response** of LTI system. It is also called **transfer function**.

- **Magnitude and Phase Representation of Frequency Response:**

- The frequency response $H(e^{j\omega})$ can be written in polar form as

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\theta(e^{j\omega})} \quad \dots.i$$

where, $|H(e^{j\omega})|$ = amplitude (or magnitude) response

and $\theta(e^{j\omega})$ = phase response

- Note that phase response does not affect the amplitude of the individual frequency components but only provides information concerning the relative phases of exponentials that make up $h[n]$.

➤ $H(e^{j\omega})$ exhibits conjugate symmetry. That is,

$$|H(e^{j\omega})| = |H(-e^{j\omega})| \quad , \text{ symmetric about origin.}$$

and

$$\theta(-j\omega) = -\theta(j\omega) \quad , \text{ antisymmetric about origin.}$$

❑ Sampling of continuous time signal, spectral properties of sampled signal:

- Mostly, periodic and uniform sampling is used, which is given by the equation

$$x[n] = x_a(nT), \quad -\infty < n < \infty \quad \text{.....1}$$

where, $x[n]$ = discrete-time signal obtained by taking samples of the analog signal $x_a(t)$ every T seconds (i.e., $t = nT$)

- The time interval T between two successive samples is called *sampling period or sampling interval* and its reciprocal $\frac{1}{T} = F_S$ is called *sampling rate of sampling frequency*.

- Also, we can write,

$$t = nT = \frac{n}{F_S} \quad \text{.....2}$$

where, t = time variable of continuous-time signal

and, n = time variable of discrete-time signal

- We can establish the relationship between frequency variable F (*or* Ω) for analog frequency and the frequency variable f (*or* ω) for the discrete-time signal by the equation

$$x_a(t) = A \cos(2\pi F t + \theta) \quad \text{.....3}$$

$$\Omega = 2\pi F$$

- After sampling at $F_S = \frac{1}{T}$ samples/sec. (or Hz), we have

$$x_a(nT) \equiv x[n] = A \cos(2\pi F nT + \theta)$$

$$x[n] = A \cos(2\pi \frac{F}{F_S} n + \theta)$$

$$x[n] = A \cos(2\pi f n + \theta) \quad \text{.....4}$$

where, $f = \frac{F}{F_S}$ is the *relative or normalized frequency (DT signal frequency)*.

- The range of frequency variable F and Ω for continuous-time (CT) sinusoids are

$$-\infty < \mathbf{F} < \infty$$

$$-\infty < \mathbf{\Omega} < \infty$$

But for the discrete-time (DT) sinusoids are

$$-\frac{1}{2} < \mathbf{f} < \frac{1}{2}$$

$$-\pi < \mathbf{\omega} < \pi$$

Since, $f = \frac{F}{F_S}$, we can write

$$-\frac{F_S}{2} = -\frac{1}{2T} \leq \mathbf{F} \leq \frac{F_S}{2} = \frac{1}{2T}$$

$$-\pi F_S = -\frac{\pi}{T} \leq \mathbf{\Omega} \leq \pi F_S = \frac{\pi}{T}$$

- Therefore, periodic sampling of a CT signal implies a mapping of infinite frequency range for the variable $\mathbf{F}$ (**or** $\mathbf{\Omega}$) into a finite frequency range of a DT signal for the variable $\mathbf{f}$ (**or** $\mathbf{\omega}$).

- The highest frequency F_{max} that can be represented in a digital signal of a specified sampling frequency. It is equal to one-half of the sampling rate. That is,

$$F_{max} = \frac{F_s}{2} = \frac{1}{2T}$$

- **Folding Frequency (F_{fold}):**

- The folding frequency is defined by the relation

$$F_{fold} = \frac{F_s}{2}$$

- This is because we can determine the mapping of any (alias) frequency above $\frac{F_s}{2}$ into the equivalent frequency below $\frac{F_s}{2}$.

- **Note:**

- **Aliasing:**

- A CT sinusoidal signal of higher frequency acquiring the identity of a sinusoidal signal of lower frequency after sampling is called aliasing. It occurs when $F_s < 2F_{max}$.

○ **Examples:**

- 1. Consider the signal , $x_a(t) = 3\cos 100\pi t$**
 - a. Determine the minimum sampling rate required to avoid aliasing.**
 - b. Suppose that the signal is sampled at the rate $F_s = 200\text{Hz}$. What is the discrete-time signal obtained after sampling?**
 - c. Suppose that the signal is sampled at the rate $F_s = 75\text{Hz}$. What is the discrete-time signal obtained after sampling?**
 - d. What is the frequency $0 < F < \frac{F_s}{2}$ of a sinusoid that yields samples identical to those obtained in part (b) and (c)?**

Solution:

➤ Given, $x_a(t) = 3\cos 100\pi t = 3\cos 2\pi(50)t$

a. The analog signal frequency is , $F = 50\text{ Hz}$

➤ The minimum sampling rate (Nyquist rate) to avoid aliasing is

$$F_s = 2F_{max} = 2 \times 50 = 100\text{ Hz}$$

- b. Given , $F_s = 200Hz$, then the DT (discrete-time) signal is obtained by putting $t = nT = \frac{n}{F_s}$ and $f = \frac{F}{F_s}$.

$$x[n] = 3\cos 2\pi \left(\frac{50}{200}\right) n = 3\cos 2\pi \left(\frac{1}{4}\right) n = \mathbf{3\cos \frac{\pi}{2} n}$$

Here, $f = \frac{1}{4}$ samples/sec.

- b. Given , $F_s = 75 Hz$, then the DT (discrete-time) signal is obtained by putting $t = nT = \frac{n}{F_s}$ and $f = \frac{F}{F_s}$.

$$x[n] = 3\cos 2\pi \left(\frac{50}{75}\right) n = 3\cos \frac{4\pi}{3} n = 3 \cos \left(2\pi - \frac{2\pi}{3}\right) n = \mathbf{3\cos \frac{2\pi}{3} n}$$

Here, $f = \frac{1}{3}$ samples/sec.

- d. We know, the CT signal obtained from DT signal $x[n]$ is given by the equation $y_a(t) = A\cos 2\pi Ft$

i. For $F_s = 200\text{Hz}$, we know $f = \frac{F}{F_s}$ then $F = fF_s = \frac{1}{4} \times 200 = 50\text{ Hz}$. Then,

$$y_a(t) = 3 \cos(2\pi Ft) = 3 \cos 2\pi(50)t = \mathbf{3 \cos 100\pi t}$$

ii. For $F_s = 75\text{Hz}$, we know $f = \frac{F}{F_s}$ then $F = fF_s = \frac{1}{3} \times 75 = 25\text{ Hz}$. Then,

$$y_a(t) = 3 \cos(2\pi Ft) = 3 \cos 2\pi(25)t = \mathbf{3 \cos 50\pi t}$$

Hence, $F = 50\text{ Hz}$ is an alias of $F = 25\text{ Hz}$ when $F_s = 75\text{Hz}$.

- **Note:**

➤ For the signals $\mathbf{3 \cos 100\pi t}$ and $\mathbf{3 \cos 50\pi t}$, when these signals are sampled at the rate $F_s = 75\text{Hz}$, then the DT signal obtained is same, i.e., $\mathbf{3 \cos \frac{2\pi}{3}n}$. Thus, aliasing is introduced for $F_s = 75\text{ Hz}$

2. A digital communication link carries binary-coded words representing samples of an input signal $x_a(t) = 3\cos 600\pi t + 2\cos 1800\pi t$. The link is operated at $10,000 \text{ bits/s}$ and each input sample is quantized into 1024 different voltage levels.
- What are the sampling frequency and folding frequency?
 - What is the Nyquist rate for the signal $x_a(t)$?
 - What are the frequencies for the resulting discrete-time signal $x[n]$?

Solution:

➤ The total quantized voltage level given is , $L = 1024$

Then the number of bits/sample = $\log_2 L = \log_2 1024 = \frac{\log_{10} 1024}{\log_{10} 2} = 10$

a. The sampling frequency, $F_s = \frac{10,000}{10} = 1000 \text{ Hz}$

The folding frequency , $F_{fold} = \frac{F_s}{2} = \frac{1000}{2} = 500 \text{ Hz}$

b. We have , $x_a(t) = 3\cos 2\pi(300)t + 2\cos 2\pi(900)t$

The frequencies present in the signal $x_a(t)$ are $F_1 = 300 \text{ Hz}$, $F_2 = 900 \text{ Hz}$

Therefore, $F_{max} = 900 \text{ Hz}$

now, Nyquist sampling rate, $F_N = 2F_{max} = 2 \times 900 = 1800 \text{ Hz}$

c. We have, $F_s = 1000 \text{ Hz}$. Then,

$$x[n] = 3\cos 2\pi \left(\frac{300}{1000} \right) n + 2\cos 2\pi \left(\frac{900}{1000} \right) n$$

$$x[n] = 3\cos 2\pi \left(\frac{3}{10} \right) n + 2\cos 2\pi \left(\frac{9}{10} \right) n$$

$$x[n] = 3\cos 2\pi \left(\frac{3}{10} \right) n + 2\cos \left(\frac{9\pi}{5} \right) n$$

Or,
$$x[n] = 3\cos 2\pi \left(\frac{3}{10}\right)n + 2\cos \left(2\pi - \frac{\pi}{5}\right)n$$

$$x[n] = 3\cos 2\pi \left(\frac{3}{10}\right)n + 2\cos 2\pi \left(\frac{1}{10}\right)n \quad \text{.....i}$$

$$x[n] = 3\cos \frac{3\pi}{5}n + 2\cos \frac{\pi}{5}n \quad \text{.....ii}$$

➤ Therefore, the frequencies in DT signal $x[n]$ are

$$f_1 = \frac{3}{10} \text{ Hz} \quad \text{and} \quad f_2 = \frac{1}{10} \text{ Hz}$$

d. Resolution,
$$\Delta = \frac{x_{\max} - x_{\min}}{M-1} = \frac{5 - (-5)}{1024-1} = \frac{10}{1023}$$

- **Note:**

i. $x_{\max} - x_{\min}$ is called dynamic range .

ii. Resolution is also known as Quantization Step Size

3. Given the analog signal, $x(t) = 2 \sin^2 300\pi t + 5 \sin 500\pi t$

- i. What is the Nyquist rate of sampling?**
- ii. If the given signal is sampled at a rate of 5000 samples/s , what is the discrete-time signal obtained?**
- iii. What is the folding frequency? (Assignment)**

Solution: